

Filter

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SDU Robotics

Topics to be covered in this course

- Sampling and reconstruction
- Aliasing
- Quantization and dynamic range
- Implementation
- Conversion time-frequency domain
- Z transform
- Linear Time Invariant system (LTI)
- System analysis
- Window functions
- Filter design
- Impulse response (FIR and IIR)

Digital Signal Processing Systems,
Basic Filtering Types, and Digital
Filter Realizations

Reference book

Chapter 6:

6.5 Basic Types of filtering

6.6 Realization of Digital filters

(Page 186-199)

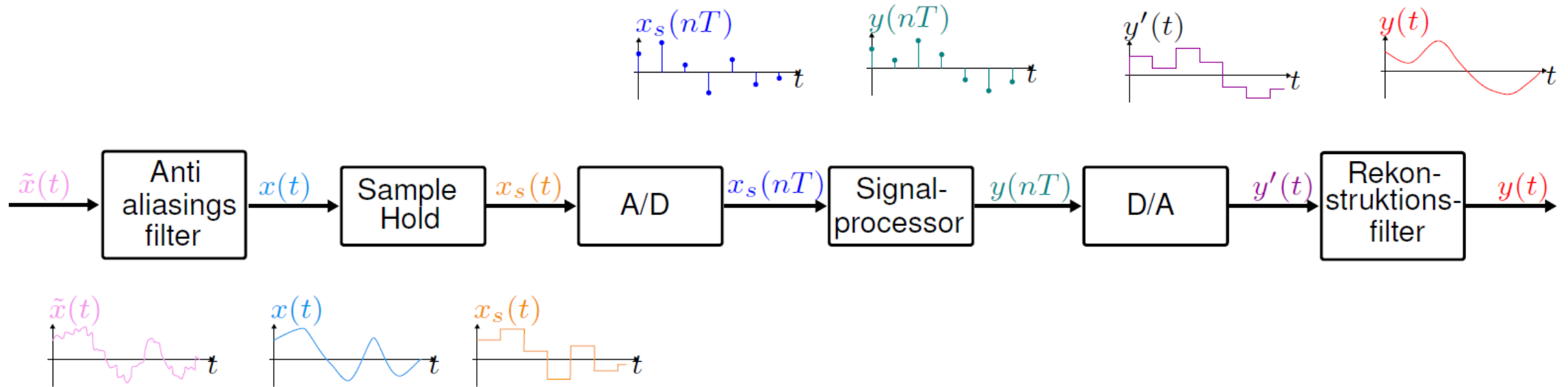
Infinite Impulse Response
Filter Design

Reference book

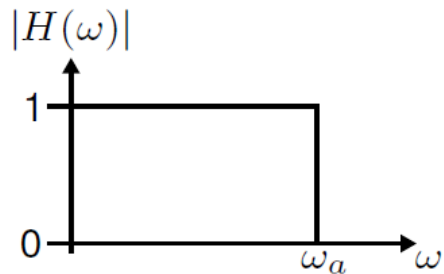
Chapter 8:

8.3 Digital Butterworth and Chebyshev Filter Designs
(Page 318-338)

Digital signal processing workflow

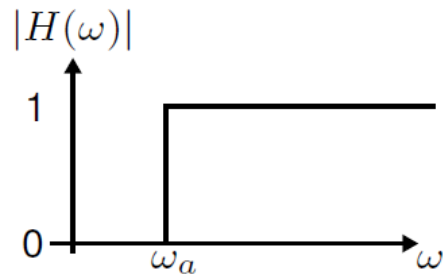


Filters



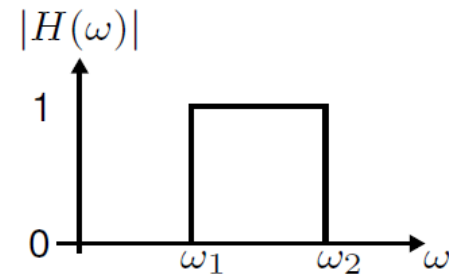
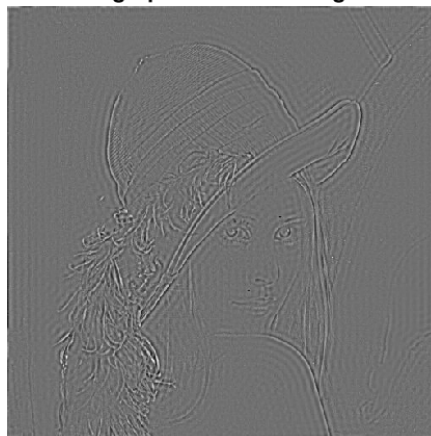
Low pass

Low-pass Filtered Image



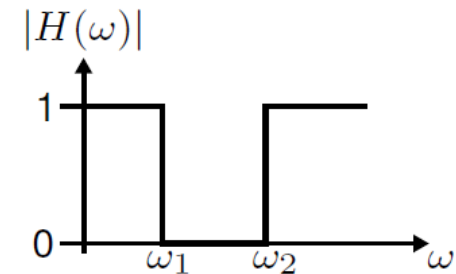
High pass

High-pass Filtered Image



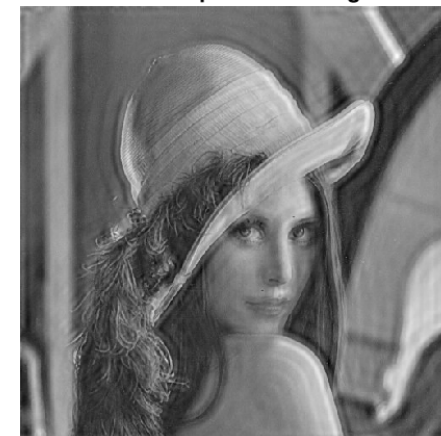
Band pass

Band-pass Filtered Image



Band stop

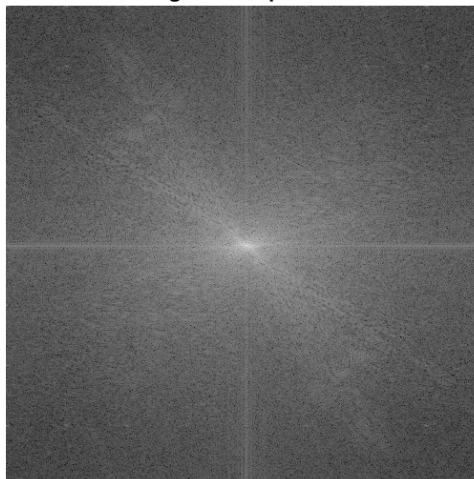
Band-stop Filtered Image



Original Image



Magnitude Spectrum

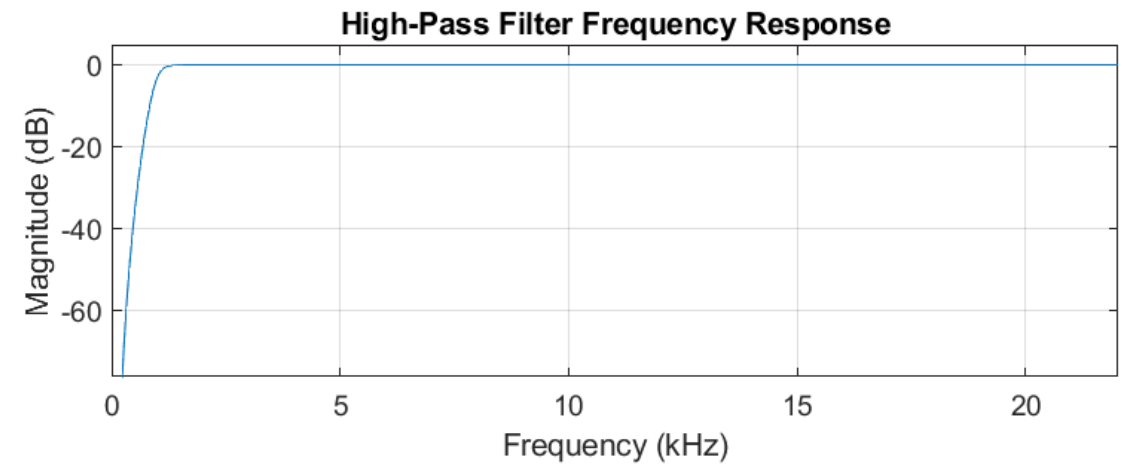
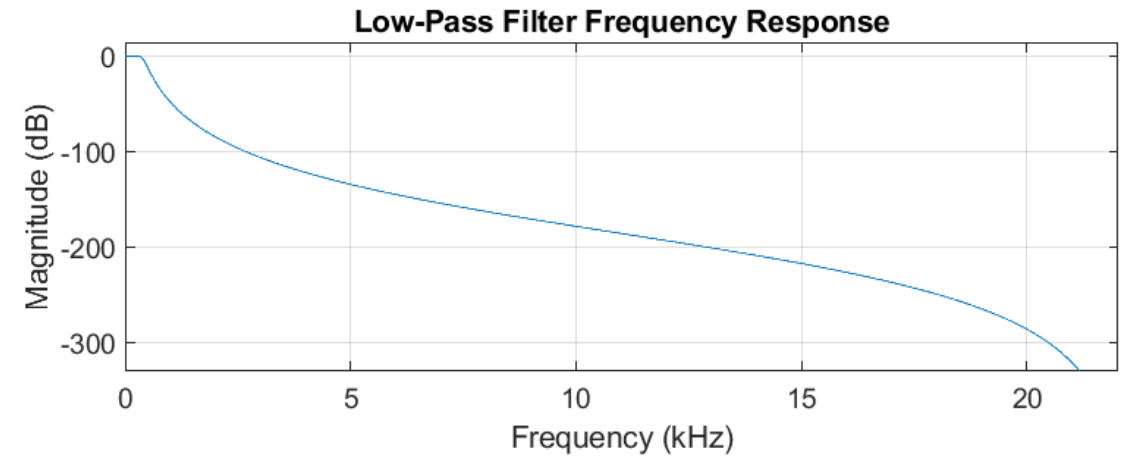
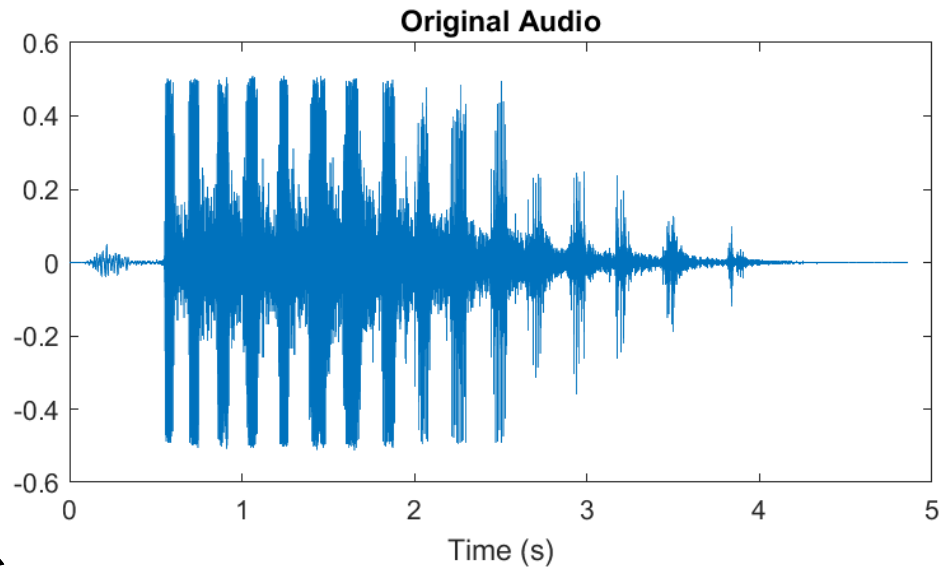


Sound signal processing

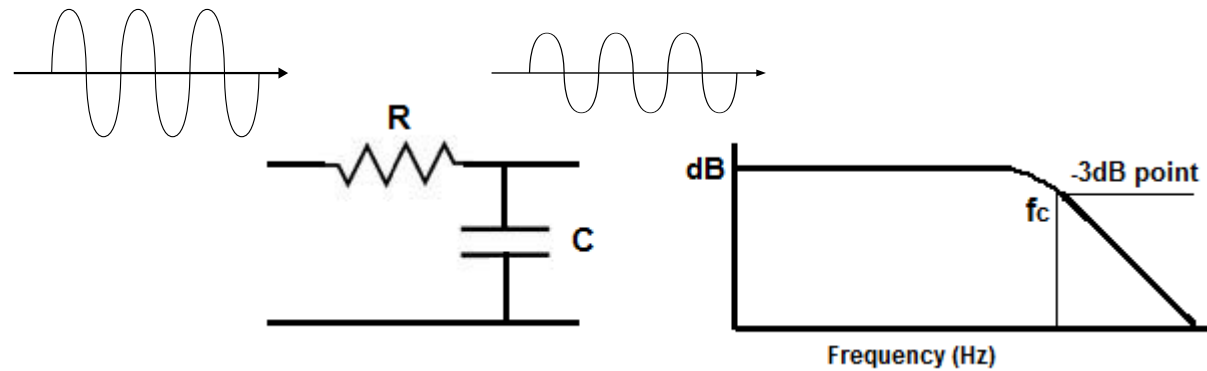
→ Original soundtrack

→ Low frequency soundtrack

→ High frequency soundtrack



Analog low pass filter



$$f_c = \frac{1}{2\pi RC}$$

For analog signal, physical components are chosen to construct an analog filter. (**Hardware**)

V_s is the source voltage (input). V_o is the output voltage on the resistor.
 i is the current.

$$V_s = Ri + L \frac{di}{dt} + \frac{1}{C} \int i dt$$

$$\frac{i}{V_s} = \frac{1}{R + sL + (sC)^{-1}}$$

So

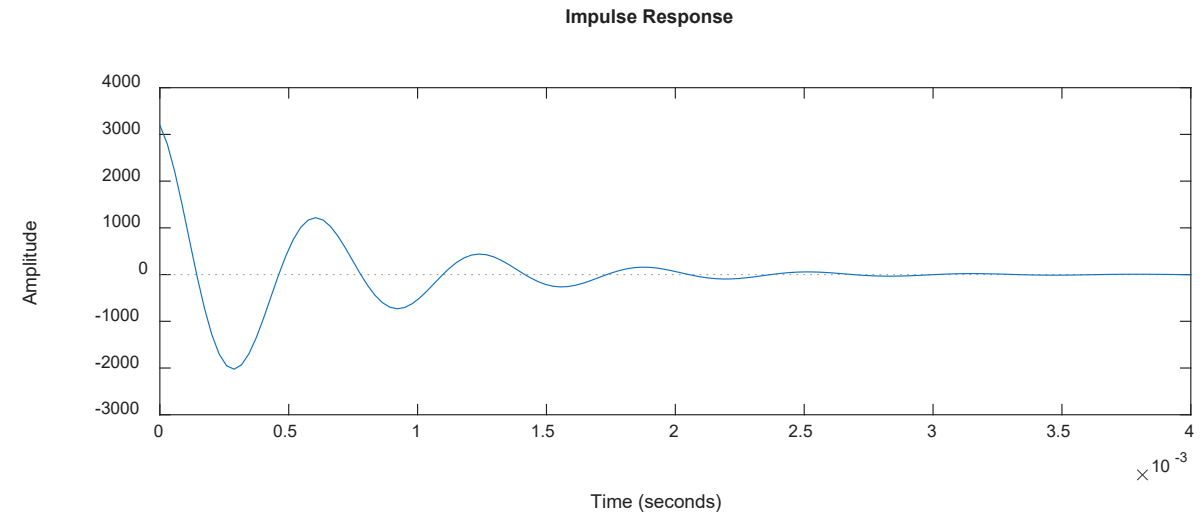
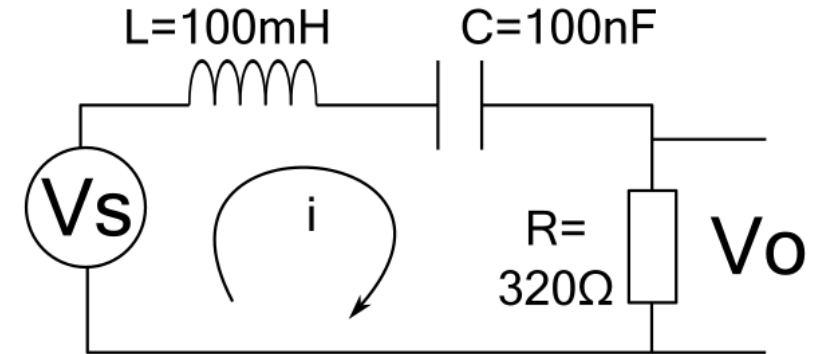
$$H(s) = \frac{V_o}{V_s} = \frac{iR}{V_s} = \frac{3200s}{s^2 + 3200s + 10^8}$$

If input $V_s = \delta(0)$, $X(s) = 1$

$$Y(s) = V_o = \frac{3200s}{s^2 + 3200s + 10^8}$$

#Matlab:

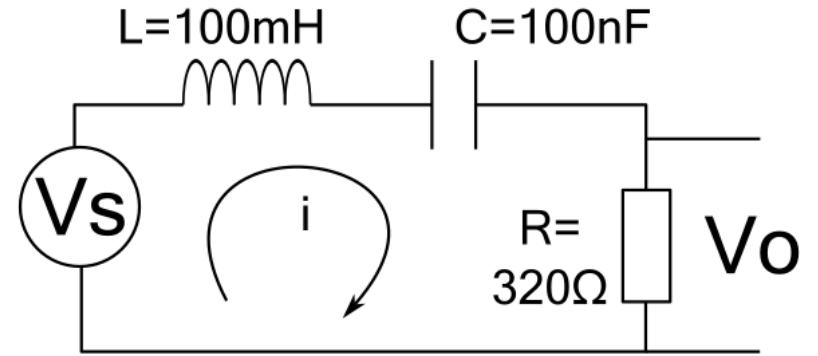
```
sys=tf([3200,0],[1, 3200,10^8]);
impulse(sys);
```



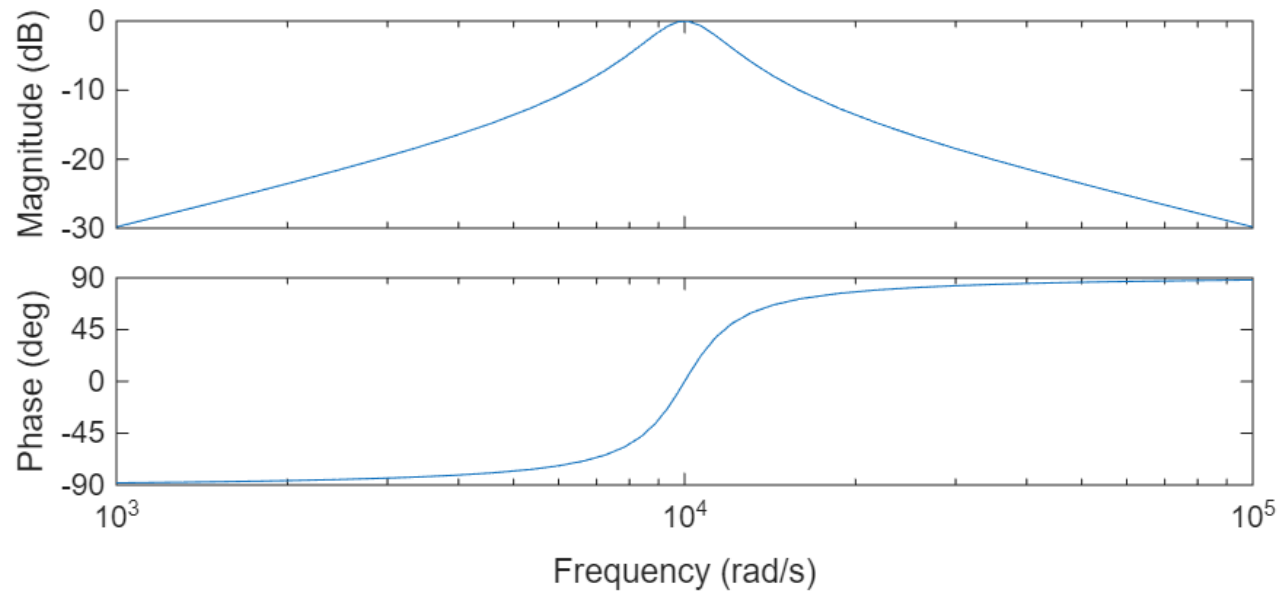
You can also test its frequency response.

$$H(s) = \frac{3200s}{s^2 + 3200s + 10^8}$$

```
#Matlab:  
sys=tf([3200,0],[1, 3200,10^8]);  
bode(sys);
```



Bode Diagram



What type of filter is it?

- Highpass
- Lowpass
- Bandpass
- Bandstop

Digital signal and digital filter

Digital signal is a type of signal that **represents information using discrete values**.

Example of digital signal:

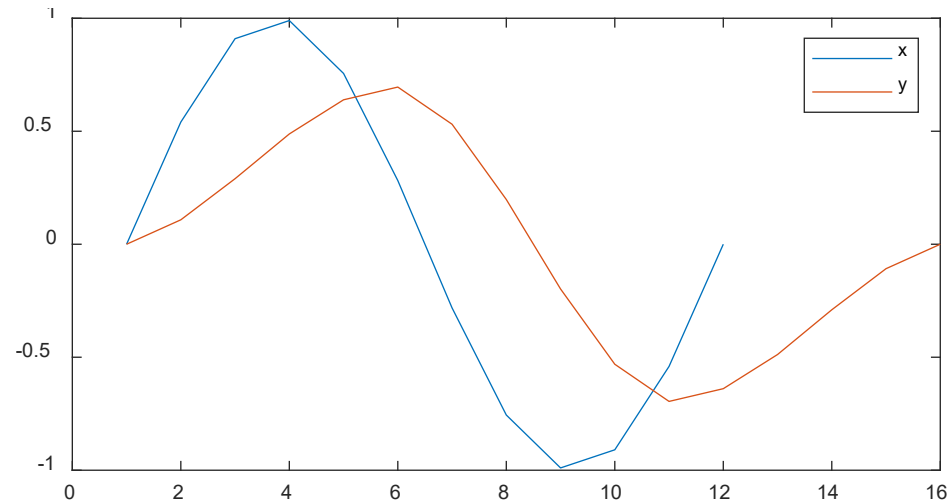
$$x(n) = [0, 0.541, 0.910, 0.990, 0.756, 0.282, -0.282, -0.756, -0.990, -0.910, -0.541, 0]$$

Example of digital low pass filter (moving average):

$$h = [0.2, 0.2, 0.2, 0.2, 0.2]$$

Operation: convolution

$$y(n) = x(n) * h$$



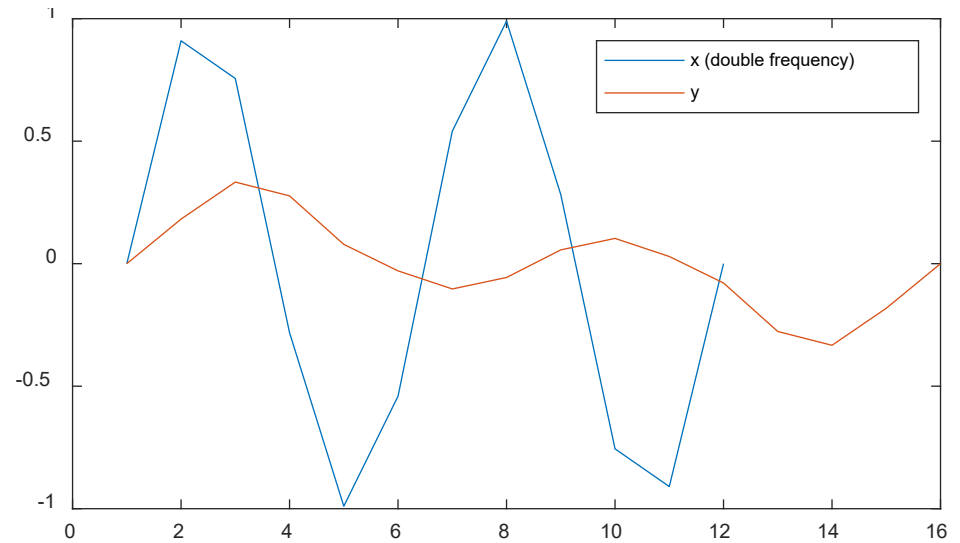
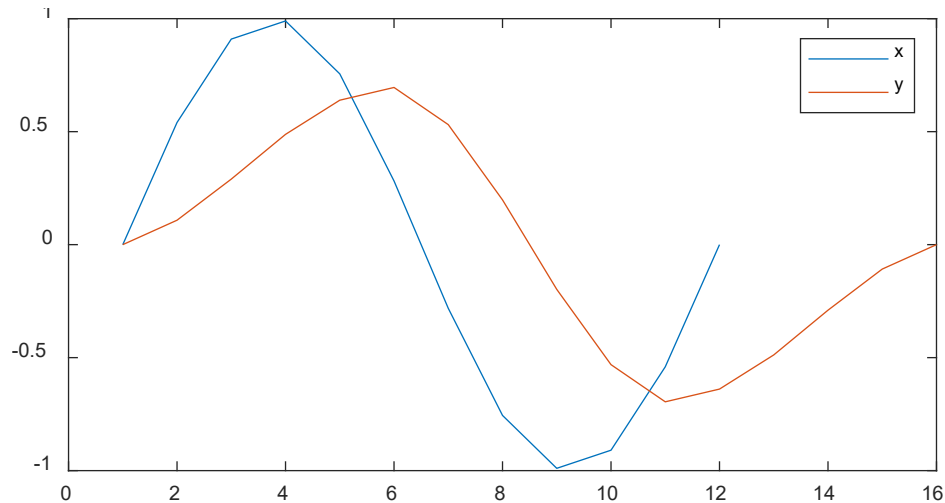
Digital signal and digital filter

If the input digital signal x becomes double frequency, and the same digital filter (moving average) is used:

$$h = [0.2, 0.2, 0.2, 0.2, 0.2]$$

Operation: convolution

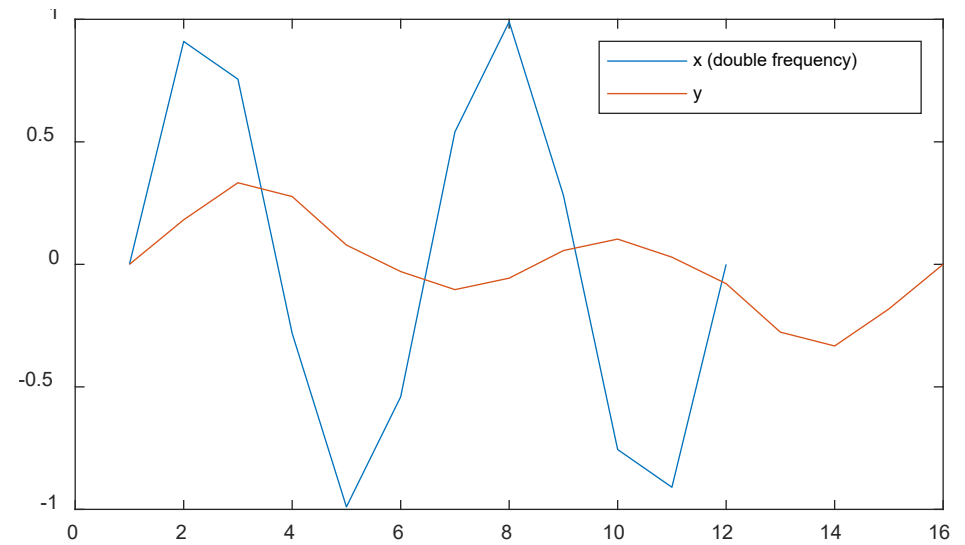
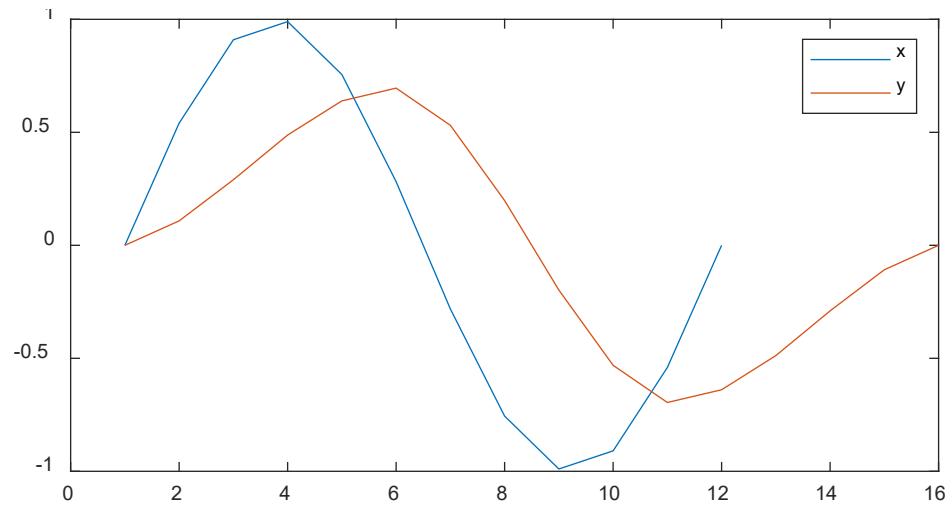
$$y(n) = x(n) * h$$



Output amplitude reduce!
Because H is a low pass filter!

Group delay

You also observe a delay?



Group Delay

For most application, the magnitude is the main concern.

But the **phase characteristic** of a filter also influences the input-output behavior of the filter.

The group delay T_g is a measure of the time delay through the filter. This value often depends on the frequency ω .

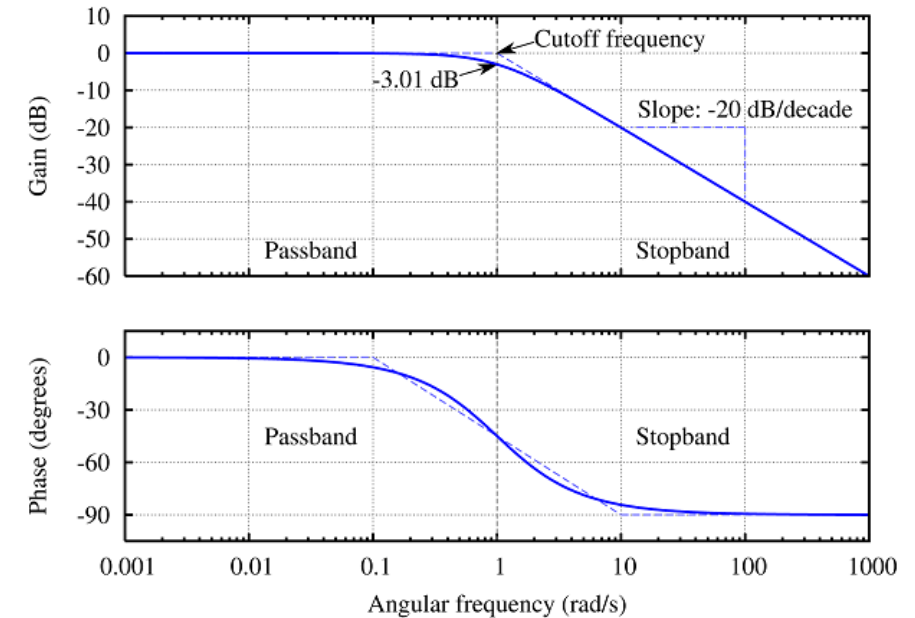
Group delay is defined as

$$T_g = -\frac{d\phi(\omega)}{d\omega}$$

where $\phi(\omega)$ is the filter phase [°].

To avoid pulse overshoot or damped oscillation, the filter must have a constant group delay.

We are talking about frequency domain here!

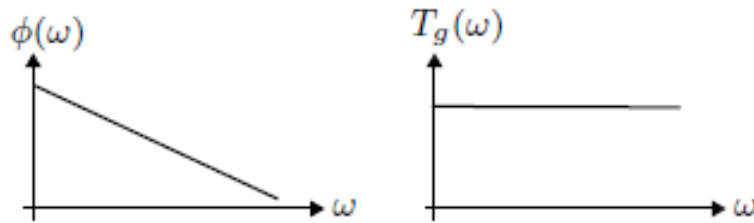


Group delay (Linear phase)

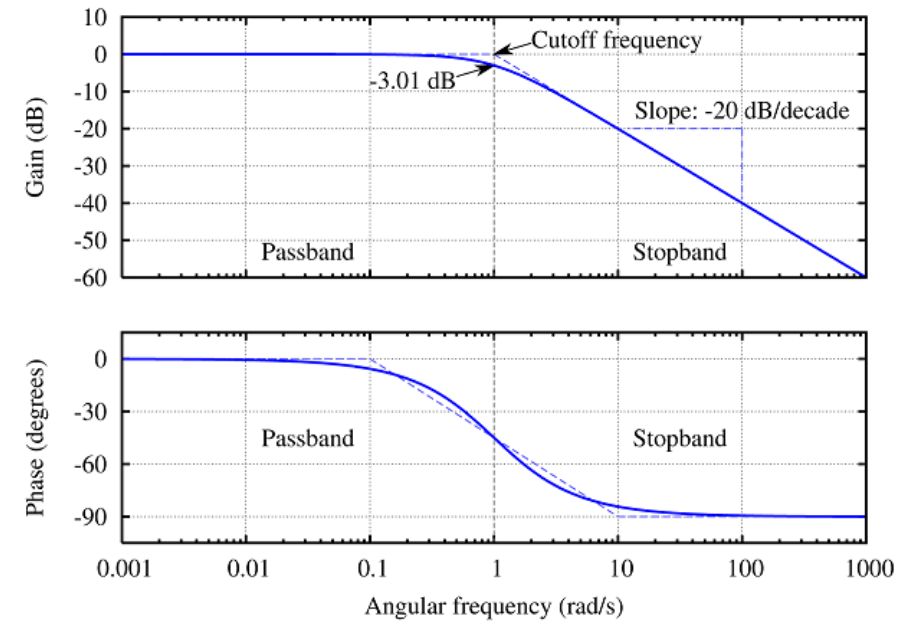
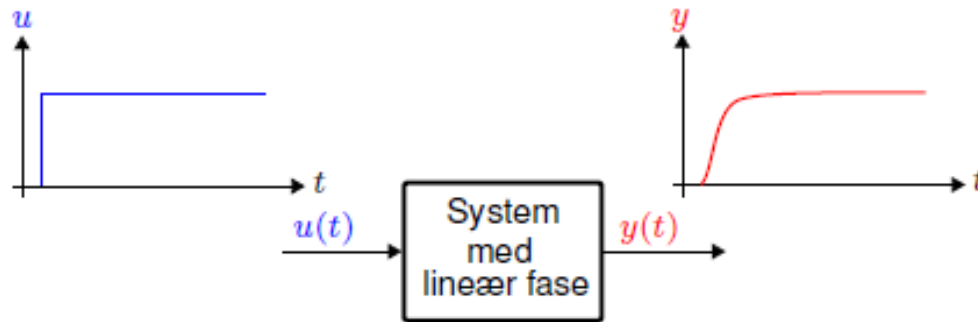
From the group delay, it can be determined whether the filter's step response has damped oscillation.

Let's consider a system with linear phase (constant group delay).

$$T_g = -\frac{d\phi(\omega)}{d\omega}$$

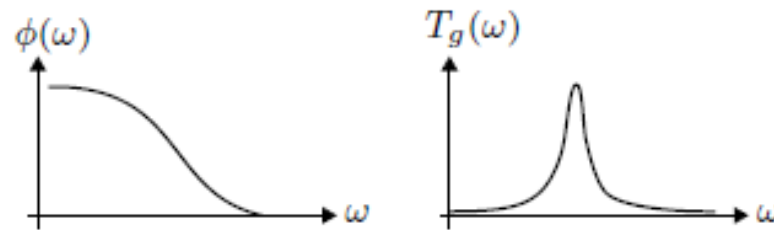


The step response of the system with linear phase does not have oscillation.

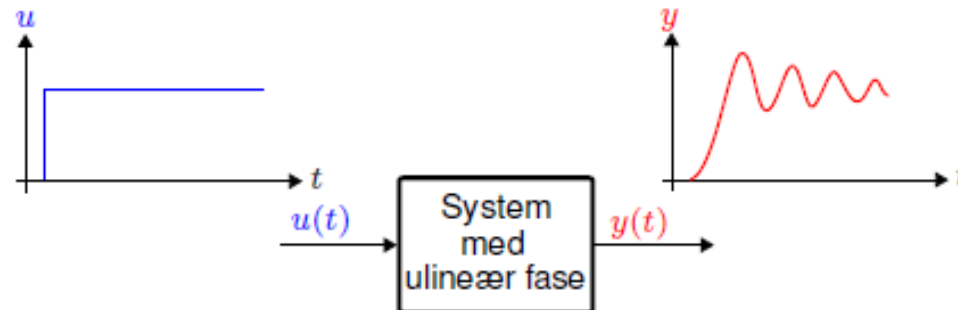


Group delay (non-linear phase)

For a system with non-linear phase



The step response of the system with nonlinear phase has oscillation.



Example – Group delay

```
% original signal
t = 0:.01:3;
% delay
tau_linear = 0;
tau_nonlinear = 0;

% different frequency component
comp1 = sin(2*pi*(t+tau_linear)+tau_nonlinear);
comp2 = sin(3*2*pi*(t+tau_linear)+tau_nonlinear)/3;
comp3 = sin(5*2*pi*(t+tau_linear)+tau_nonlinear)/5;
comp4 = sin(7*2*pi*(t+tau_linear)+tau_nonlinear)/7;
comp5 = sin(9*2*pi*(t+tau_linear)+tau_nonlinear)/9;
y = comp1 + comp2 + comp3 + comp4 + comp5;
plot(t,y);
hold on
```

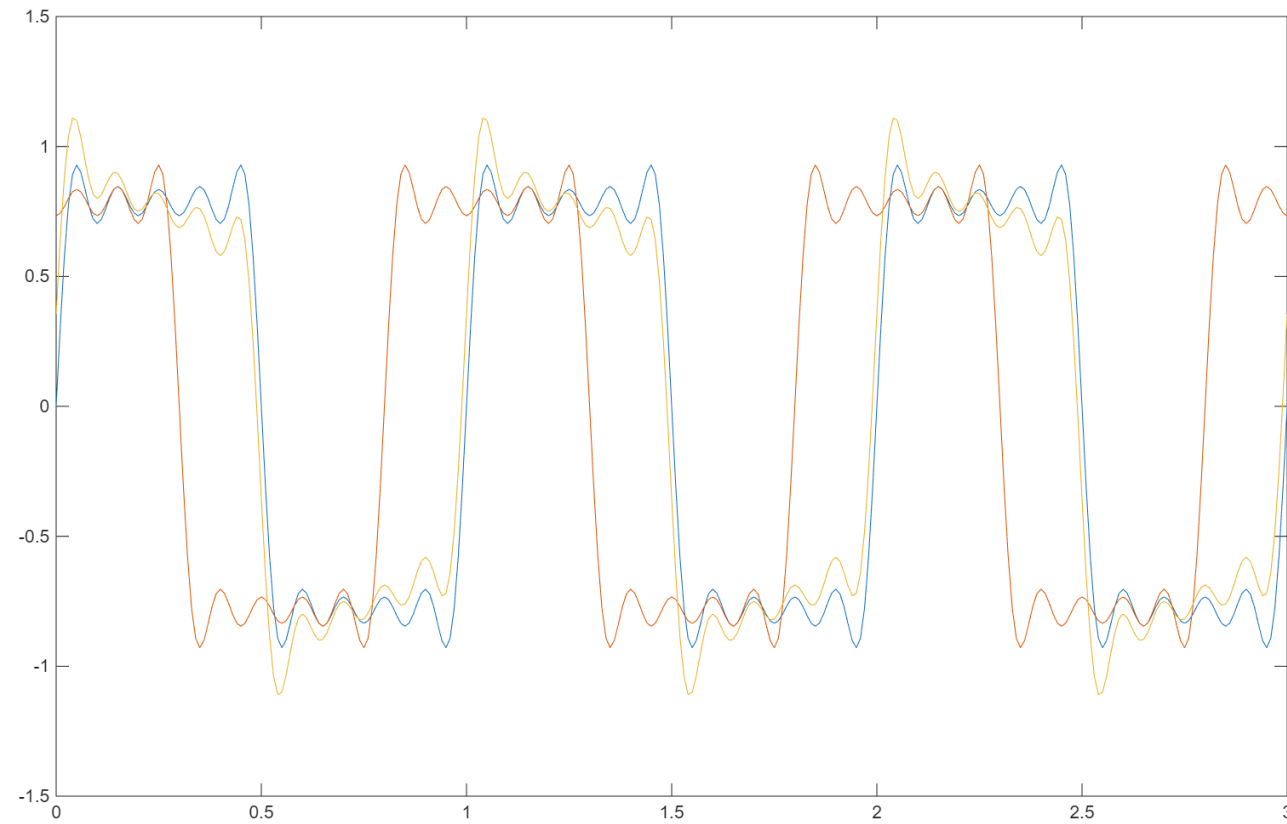
```
% linear group delay
tau_linear = 0.2;
tau_nonlinear = 0;

% different frequency component
comp1 = sin(2*pi*(t+tau_linear)+tau_nonlinear);
comp2 =
sin(3*2*pi*(t+tau_linear)+tau_nonlinear)/3;
comp3 =
sin(5*2*pi*(t+tau_linear)+tau_nonlinear)/5;
comp4 =
sin(7*2*pi*(t+tau_linear)+tau_nonlinear)/7;
comp5 =
sin(9*2*pi*(t+tau_linear)+tau_nonlinear)/9;
y = comp1 + comp2 + comp3 + comp4 + comp5;
plot(t,y);
```

```
% non-linear group delay
tau_linear = 0;
tau_nonlinear = 0.2;

% different frequency component
comp1 =
sin(2*pi*(t+tau_linear)+tau_nonlinear);
comp2 =
sin(3*2*pi*(t+tau_linear)+tau_nonlinear)/3;
comp3 =
sin(5*2*pi*(t+tau_linear)+tau_nonlinear)/5;
comp4 =
sin(7*2*pi*(t+tau_linear)+tau_nonlinear)/7;
comp5 =
sin(9*2*pi*(t+tau_linear)+tau_nonlinear)/9;
y = comp1 + comp2 + comp3 +
comp4 + comp5;
plot(t,y);
```


Group delay



Filter specification

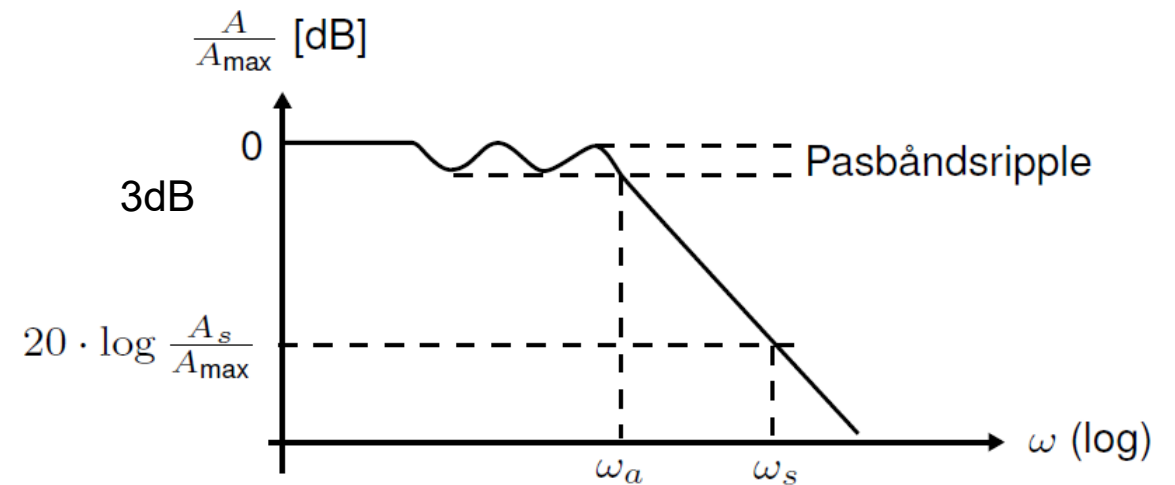
A filter specification consists of minimum requirements for the amplitude characteristic of the filter (there may also be requirements for the phase).

For a low-pass filter, these requirements can be the followings:

1. A **cutoff frequency** ω_a which indicates the upper limit of the passband.
2. A **stopband frequency** ω_s at which a **stopband attenuation** A_s is specified.
3. Information on permissible gain variation in the passband.

$$dB = 10 \log_{10} \frac{P_2}{P_1}$$
$$dB = 20 \log_{10} \frac{A_2}{A_1}$$

P denotes signal power.

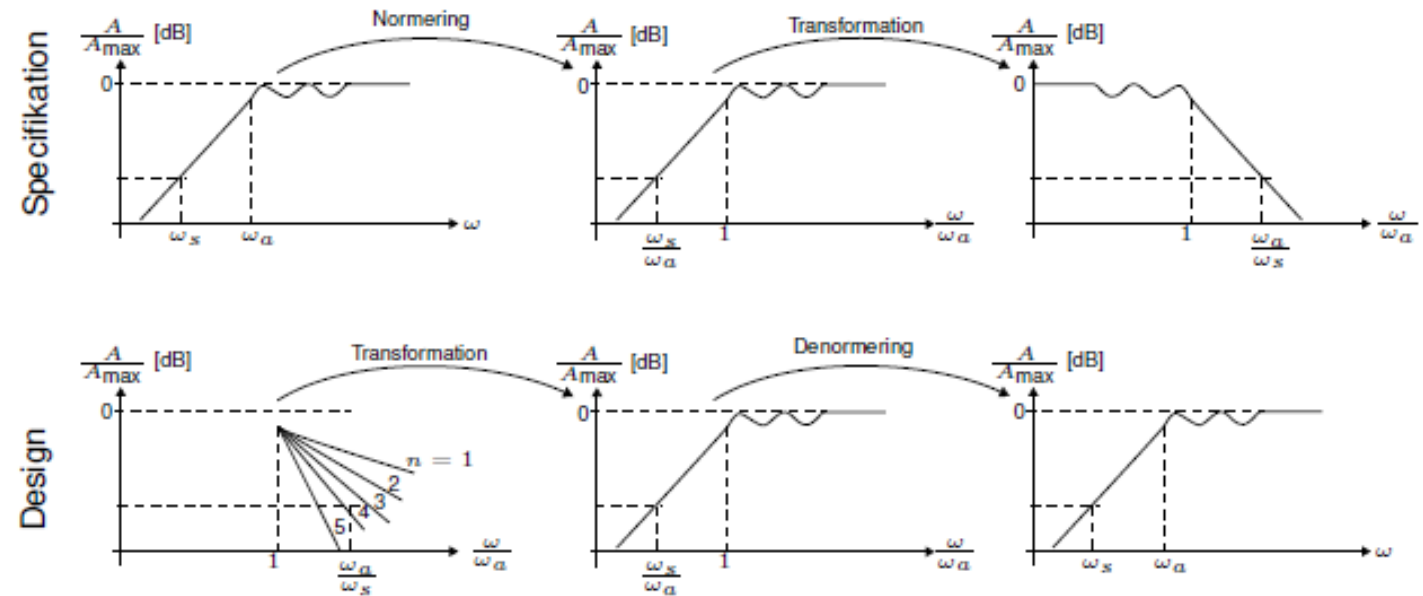


Two ways to design a filter

Method 1

Overview of the design procedure:

- Choose a filter function type
- Generate a normalized filter function
- Choose the filter order
- Convert it to the required type (Lowpass, highpass, bandpass or bandstop)
- Convert it to the required cut-off frequency



Normalized filter transfer function

In practice, the ideal filters cannot be realized, but can be approximated by using filter transfer functions.

$$H = \frac{Y}{X}$$

→ Low-pass filter

→ Cut-off frequency $\omega_a = 1$

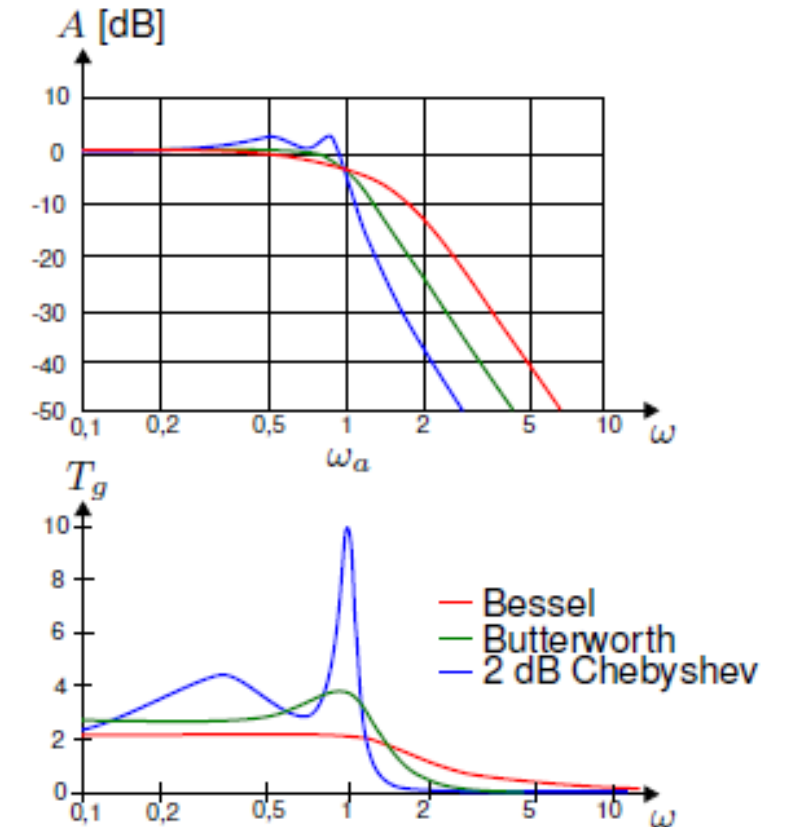
Filter transfer function types

There are three typical filter function types:

1. Butterworth
2. Chebyshev
3. Elliptic
4. Bessel

When choosing the filter function (types) for a given application, we are interested in the following characteristics:

1. Constant gain in the passband
2. High attenuation after the cutoff frequency
3. Linear phase



Normalized Butterworth Filter

The amplitude response of a N^{th} order Butterworth filter is

$$H(s) = \frac{1}{\sum_{i=0}^n a_i s^i}$$

Specifically,

$$|H(j\omega)| = \frac{1}{\sqrt{1 + (\omega/\omega_a)^{2N}}}$$

where ω_a is the cutoff frequency [rad/s].

Butterworth Filter

A Butterworth filter has the following properties:

1. Optimal in terms of **constant gain in the passband**.
2. Has attenuation of 3 dB at the cutoff frequency and then the filter's gain drops rapidly by 20 dB/decade.
3. The **phase** of the filter is **not constant** in the passband, which causes ripple at step input.

Normalized Chebyshev filter type 1

The N^{th} order Chebyshev Filter is

$$H(s) = \frac{K}{\prod_{k=1}^N [s + \sinh(\alpha) \sin \theta_k - j \cosh(\alpha) \cos \theta_k]}$$

where $\theta_k = \frac{(2k-1)\pi}{2N}$

Its defining magnitude response is

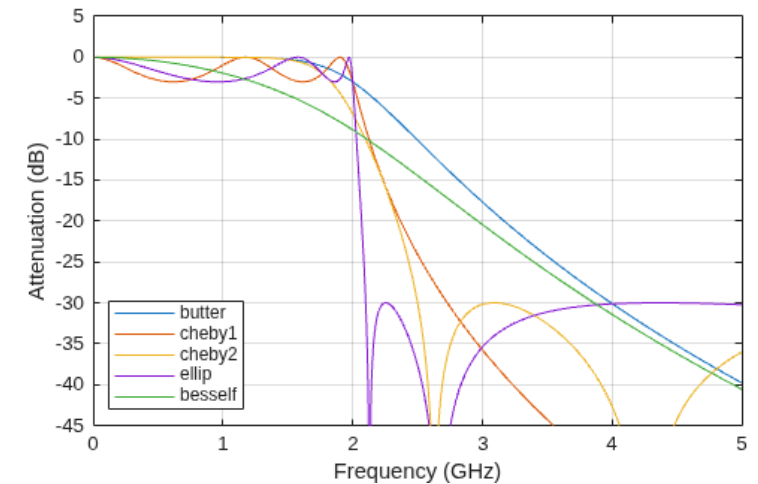
$$|H(\omega)|^2 = \frac{1}{1 + \epsilon^2 T_N^2(\omega/\omega_a)}$$

where $T_N(\omega)$ is the Chebyshev polynomial of degree N given by

$$T_N(\omega) = \begin{cases} \cos(N \cos^{-1}(\omega)) & |\omega| \leq 1 \\ \cosh(N \cosh^{-1}(\omega)) & |\omega| > 1 \end{cases}$$

$$\epsilon = \sqrt{10^{R_p/10} - 1}$$

where R_p is the size of the passband ripple in dB



Normalized Chebyshev filter type 2

The N^{th} order Chebyshev Filter is

$$H(s) = K \frac{\prod_m (s - z_m)}{\prod_{k=1}^N \left[s - \frac{1}{-\sinh(\alpha) \sin \theta_k + j \cosh(\alpha) \cos \theta_k} \right]}$$

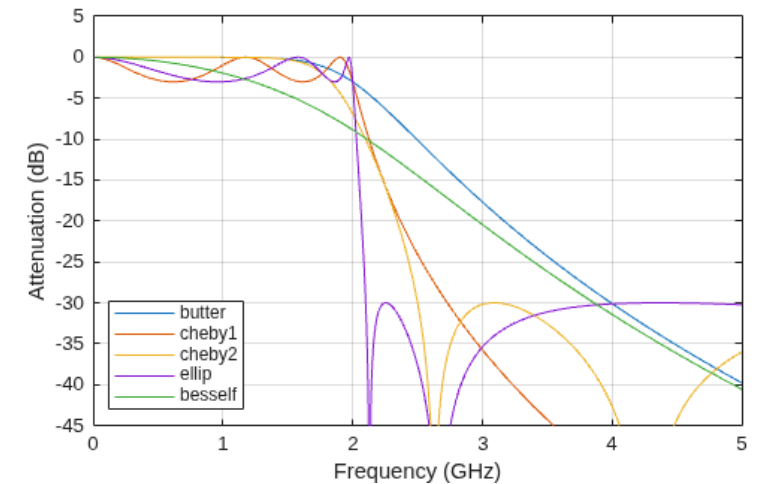
$$\text{where } \theta_k = \frac{(2k-1)\pi}{2N}$$

Its defining magnitude response is

$$|H(\omega)|^2 = \frac{1}{1 + \frac{1}{\epsilon^2 T_N^2\left(\frac{\omega_a}{\omega}\right)}}$$

where $T_N(\omega)$ is the Chebyshev polynomial of degree N given by

$$\epsilon = \sqrt{10^{R_p/10} - 1} \quad T_N(\omega) = \begin{cases} \cos(N \cos^{-1}(\omega)) & |\omega| \leq 1 \\ \cosh(N \cosh^{-1}(\omega)) & |\omega| > 1 \end{cases}$$

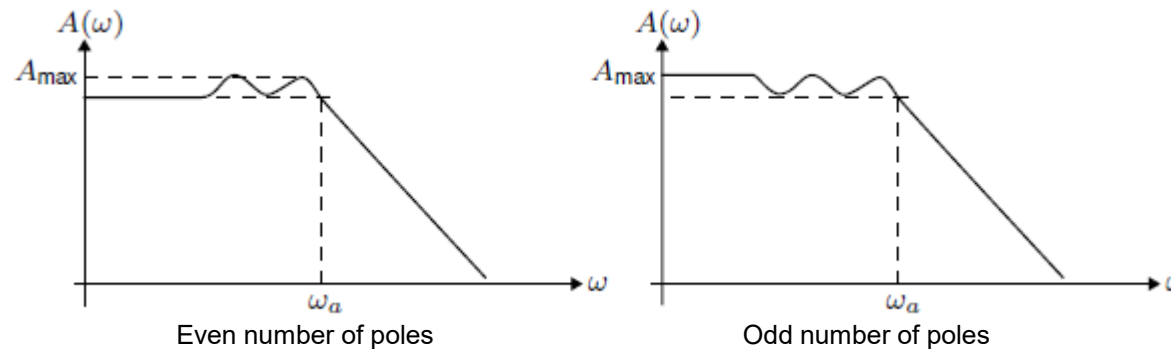


Chebyshev filter

A Chebyshev filter has the following characteristics:

1. **The size of the passband ripple can be chosen.**
2. The gain drops quickly around the cutoff frequency.
3. The **phase** of the filter is **not constant** in the passband, which leads to ripple with a step input.

The DC gain of a Chebyshev low-pass filter is not the maximum gain of the filter if the number of poles is even.



Normalized Elliptic filter type

For an even-order elliptic Filter, its transfer function is

$$H(s) = K \frac{(s^2 + \omega_{z1}^2)(s^2 + \omega_{z2}^2) \dots}{(s - p_1)(s - p_2) \dots}$$

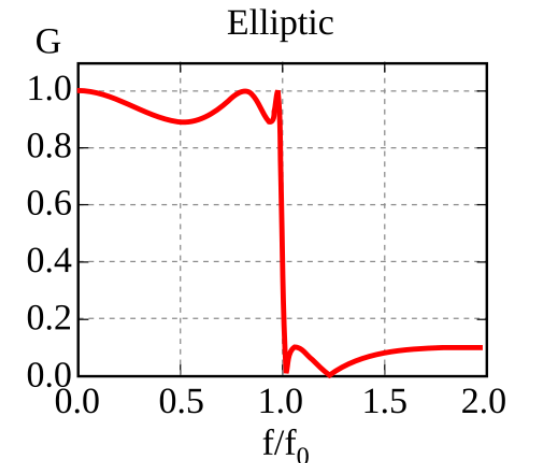
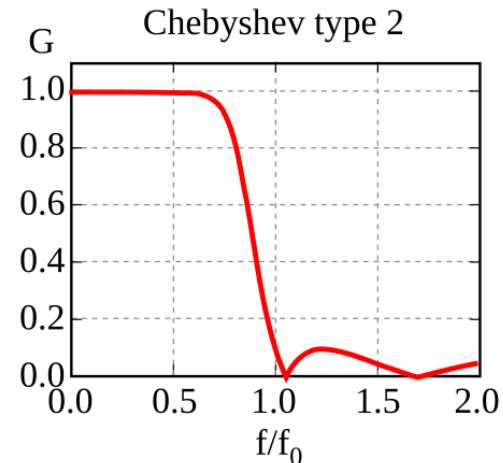
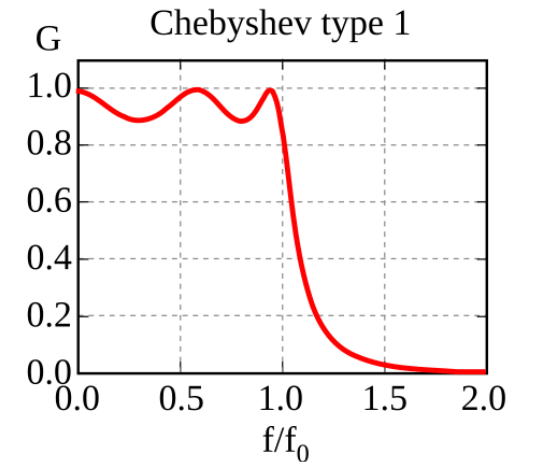
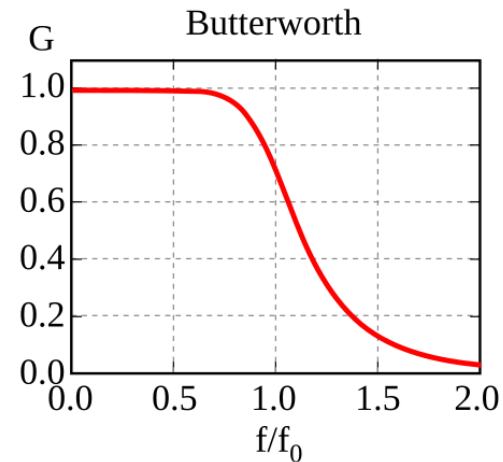
Its defining magnitude response is

$$|H(\omega)|^2 = \frac{1}{1 + \epsilon^2 R_N^2(\omega/\omega_a)}$$

You can control ripple size in passband and stopband by R_N .

Compare different filter types

	Passband	Stopband
Butterworth	monotonic	monotonic
Chebyshev I	ripple	monotonic
Chebyshev II	monotonic	ripple
Elliptic	ripple	ripple



Normalized Bessel filters

A Bessel filter has the following properties:

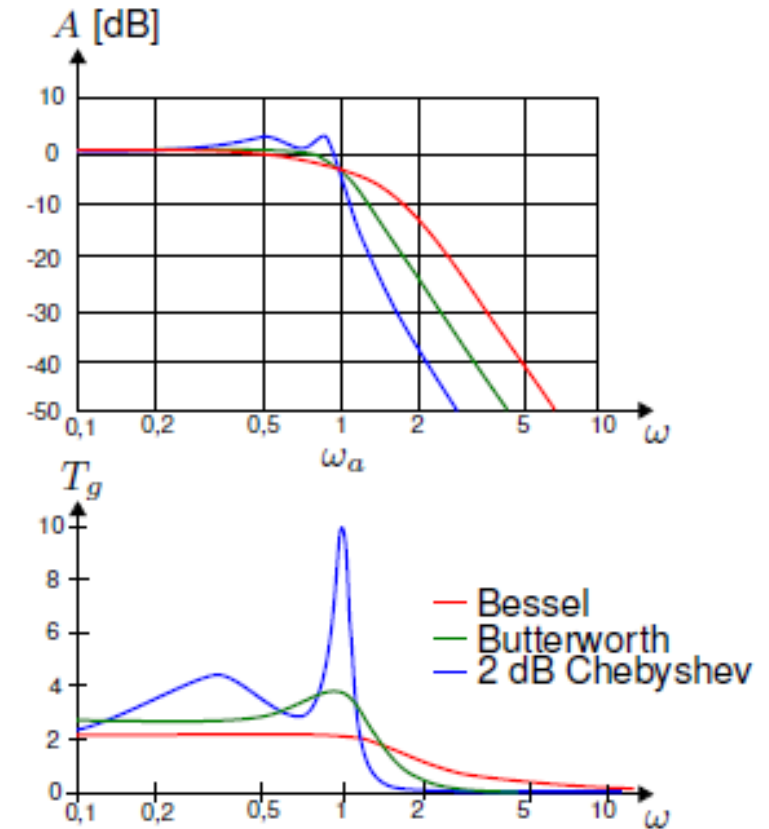
1. Has **no ripple** in the passband, but the amplitude is not as constant as with the Butterworth filter.
2. **Attenuation slowly**. The amplitude characteristic of the filter is as for the first order filter for the first 6 dB of attenuation regardless of the number of poles.
3. The **phase is almost linear** with the frequency within the passband.

The transfer function of a N^{th} order Bessel filter is

$$H_N(s) = \frac{b_0}{s^N + b_{N-1}s^{N-1} + \dots + b_1s + b_0}$$

where

$$b_k = \frac{(2N - k)!}{2^{N-k} k! (N - k)!}$$



Matlab analog filter prototype

Please check the following functions, which construct **normalized filter function**.

`[z,p,k] = buttap(n)`

`[z,p,k] = cheb1ap(n,Rp)`

`[z,p,k] = cheb2ap(n,Rp)`

`[z,p,k] = ellipap(n,Rp,Rs)`

`[z,p,k] = besslap(n)`

These functions return zeros and poles and gains.

You may need to use

```
[num,den] = zp2tf(z,p,k);
```

To convert (z,p,k) to transfer function's numerator and denominator.

Selection of the filter's order

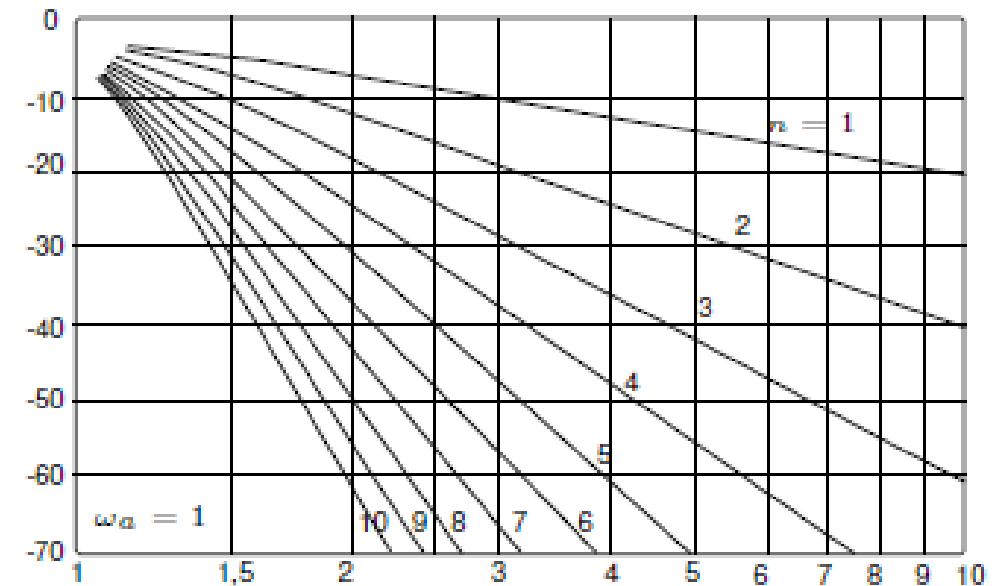
The filter's order can be found by reading the amplitude characteristic for a given filter function, and comparing this with requirements for attenuation at the stopband frequency.

The gain is normalized

$$Y = 20 \log \frac{A}{A_{\max}}$$

The frequency normalization

$$X = \frac{\omega}{\omega_a}$$



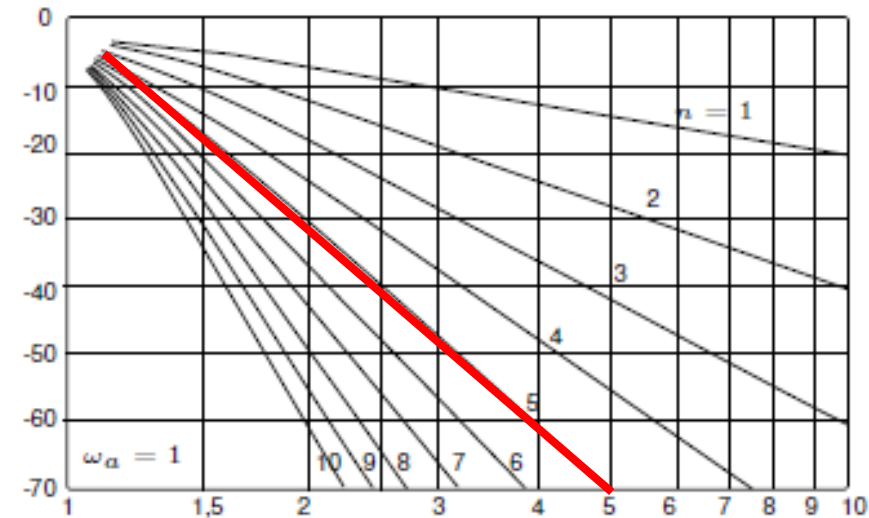
n is the filter order

Example

Find the number of poles for a low-pass filter with the requirements:

- Cutoff frequency $f_a = 2$ kHz
- Stopband frequency $f_s = 8$ kHz
- Stopband attenuation compared to A_{max} of at least 60 dB

The standard stopband frequency is $\frac{f_s}{f_a} = 4$



[num,den] = zp2tf(z,p,k);

Example

Find the number of poles for a low-pass filter with the requirements:

- Cutoff frequency $f_a = 2$ kHz
- Stopband frequency $f_s = 8$ kHz
- Stopband attenuation compared to A_{max} of at least 60 dB

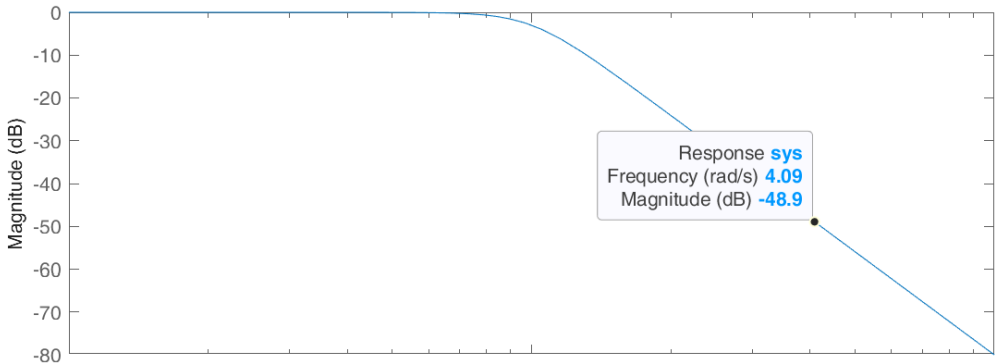
The normalized stopband frequency is $\frac{f_s}{f_a} = 4$

Then ask Matlab
For example, using Butterworth filter

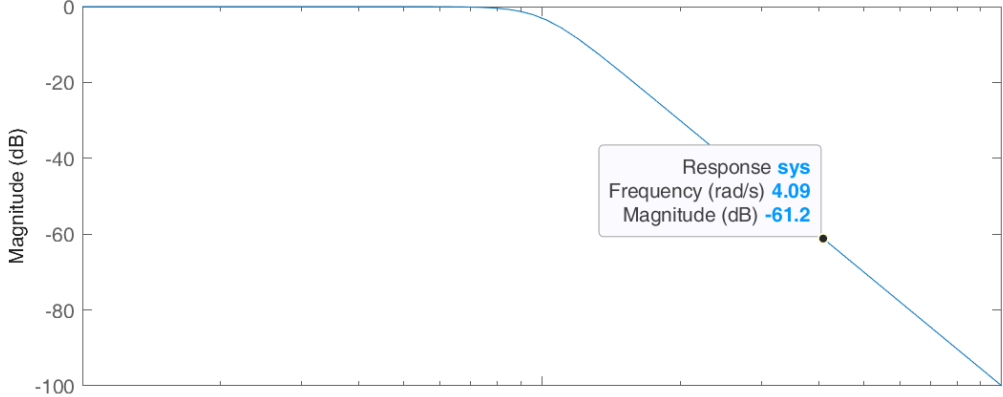
Notice: it is still an analog filter!

```
% Define filter specifications
N = 4; % Filter order (try different value)
% Butterworth Filter
[z,p,k] = buttap(N);
[num,den] = zp2tf(z,p,k);
% Check frequency response
sys = tf(num,den);
bode(sys);
```

Order of 4



Order of 5



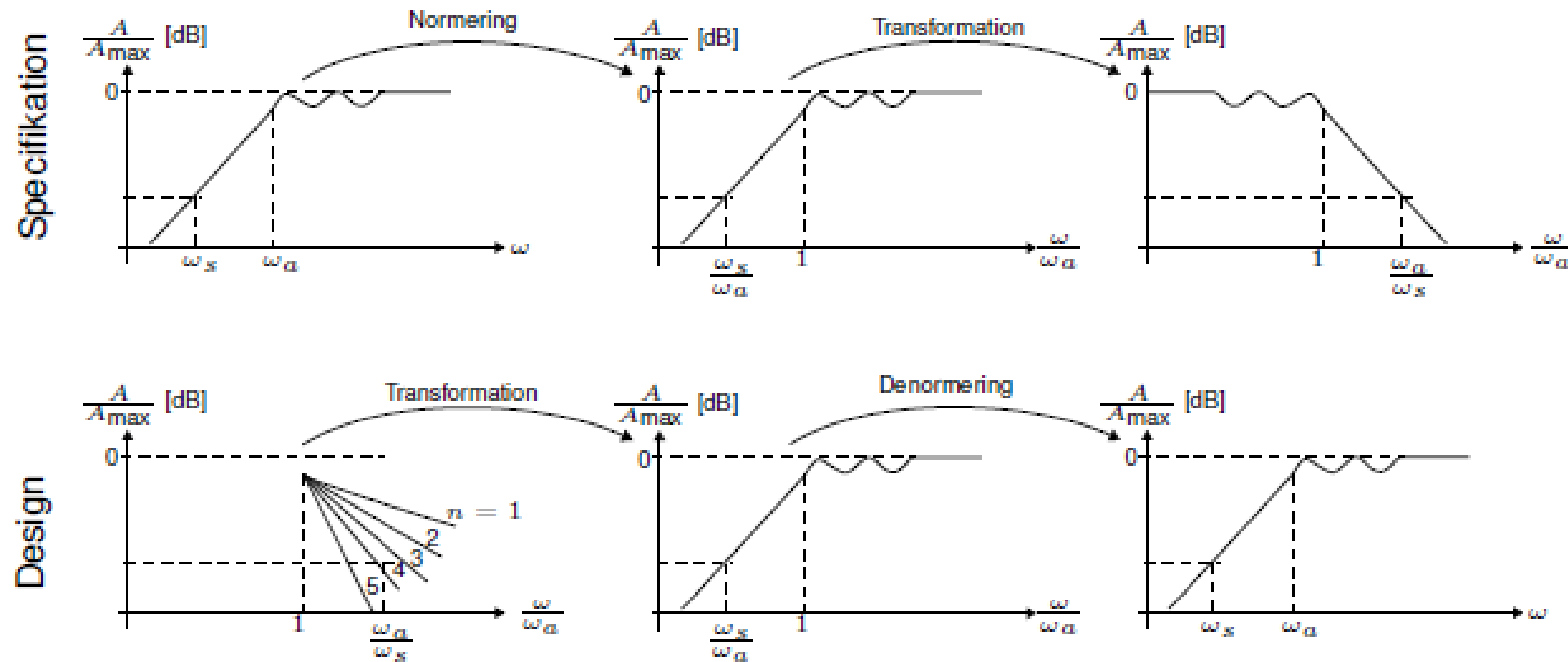
Normally, filter should be designed with less order:

- Less complexity of realization
- Less computation
- Higher precision

The transfer function can be affected by the precision limits of the system (e.g., due to rounding errors).

$$H(s) = \frac{A_0}{s^n + B_{n-1}s^{n-1} + \dots + B_1s + B_0}$$

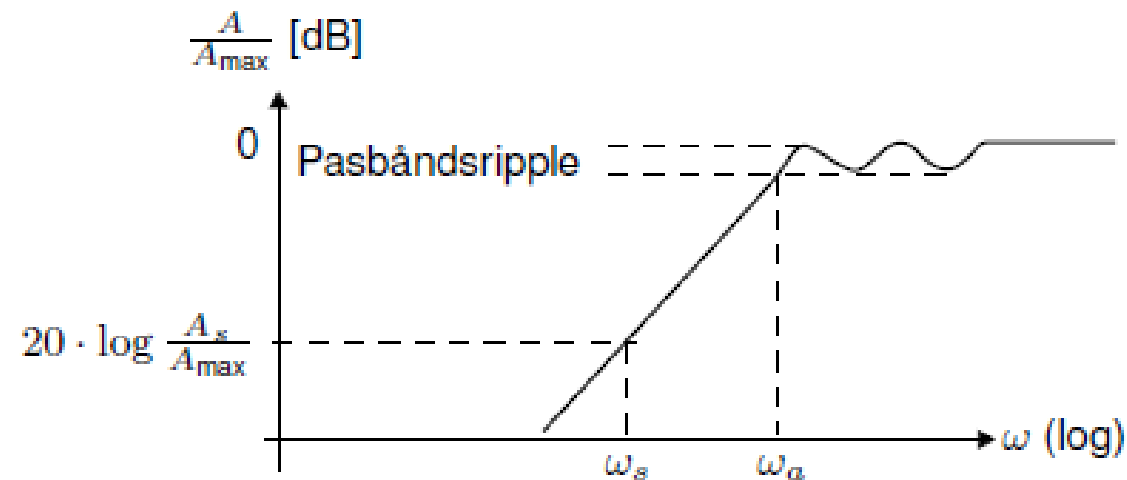
Overview of the design procedure



Low-pass to high-pass transformation

High-pass filters can be designed from normalized prototype low-pass filters based on the following specification:

- Filter function (Bessel, Butterworth, Chebyshev)
- Cutoff frequency of the filter ω_a
- Filter's stopband frequency ω_s
- Filter's stopband attenuation A_s at the stopband frequency ω_s

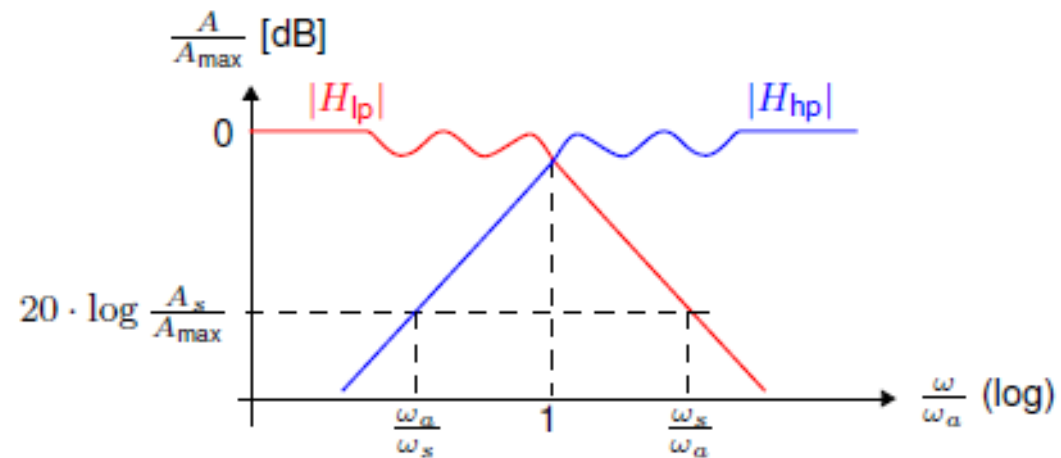


Filter transformation

A low-pass filter can be transformed into a high-pass filter by

$$H_{hp}(s) = H_{lp}(\bar{s}) \Big|_{\bar{s}=\frac{1}{s}}$$

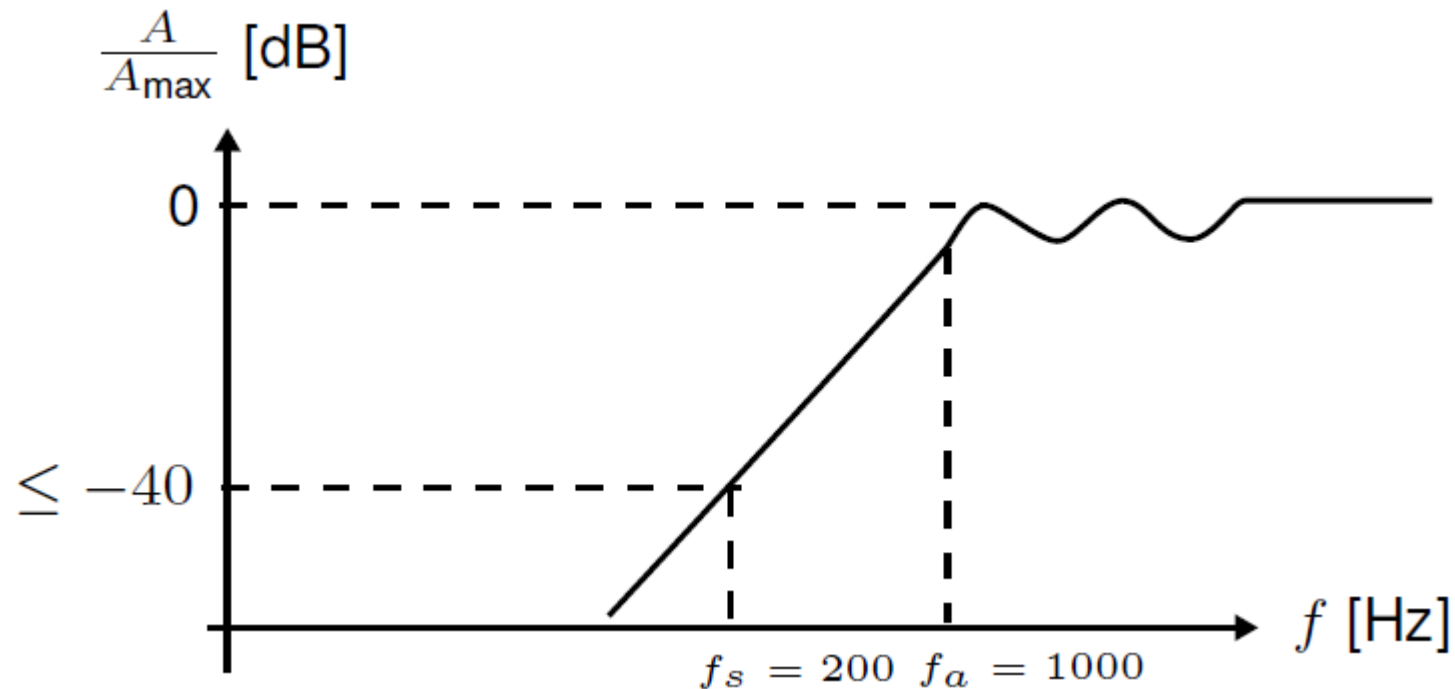
The transformation corresponds to mirroring the frequency axis by 1, as shown here



Example: Low-pass to high-pass transformation

Step 1: Filter specification

Design a 1 dB Chebyshev high-pass filter that satisfies the following amplitude characteristic.

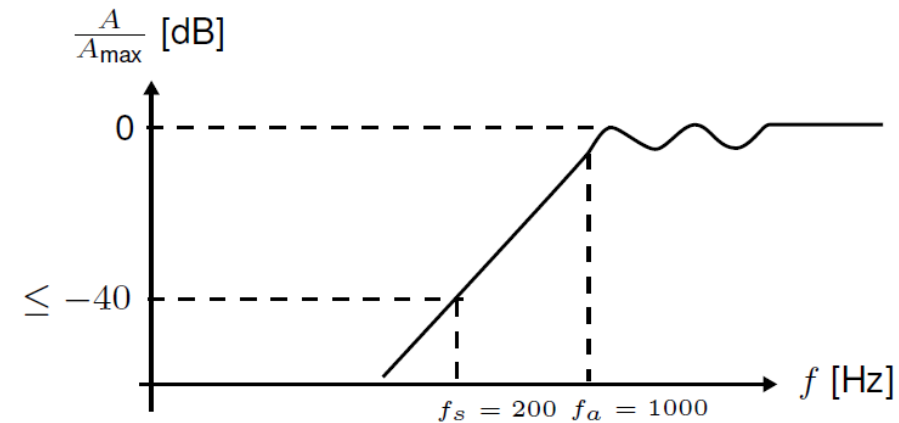


Example: Low-pass to high-pass transformation

Step 2: Transformation of specification

The stopband frequency for the **corresponding normalized low-pass filter** used for the design is calculated.

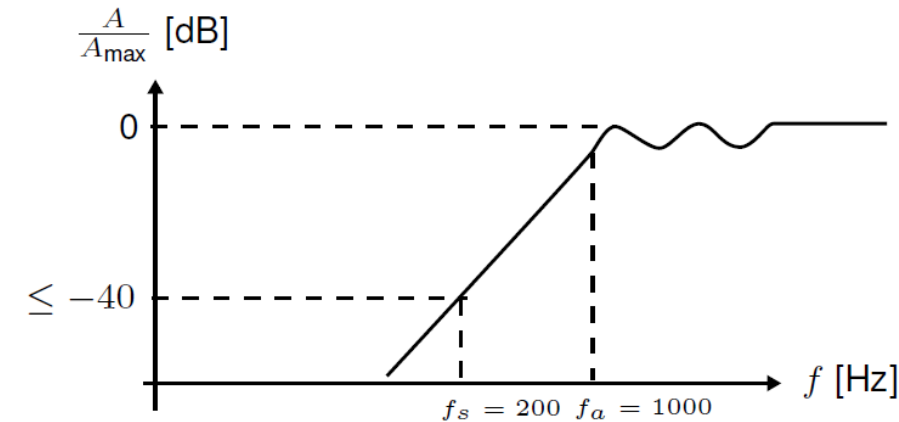
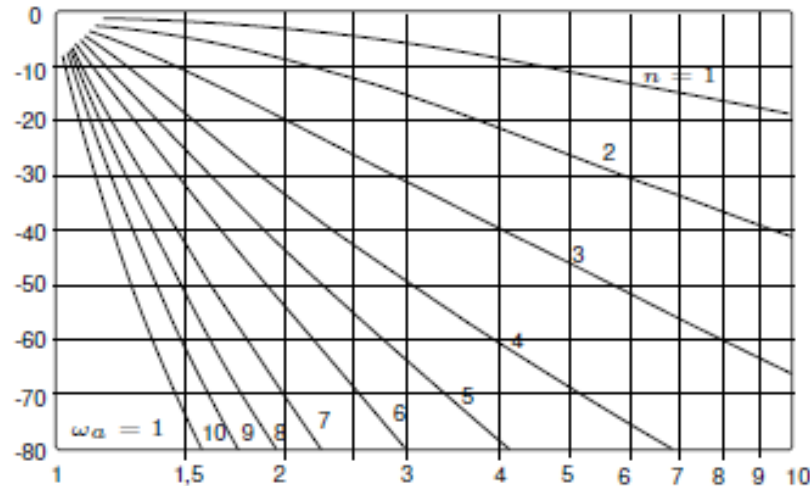
$$\frac{\omega_a}{\omega_s} = 5$$



Example: Low-pass to high-pass transformation

Step 3: Choose filter order

The filter is designed by choosing the filter's order. This is done based on the amplitude characteristics of 1 dB Chebyshev low-pass filters. It is a requirement that the gain is below **-40 dB** at a frequency of 5.



The minimum power for the filter that meets the requirement is 3 ($n = 3$).

Example: Low-pass to high-pass transformation

Step 4: Denormalization

The transfer function of the normalized **low-pass** filter can be calculated

[z,p,k] = buttap(3)

$$H(s) = \frac{1}{(s^2 + 0,49417s + 0,99421)(s + 0,49417)}$$

The normalized **high-pass** filter thus becomes (**replace s with $1/s$ in the above**)

$$H(s) = \frac{s^3}{(1 + 0,49417s + 0,99421s^2)(1 + 0,49417s)}$$

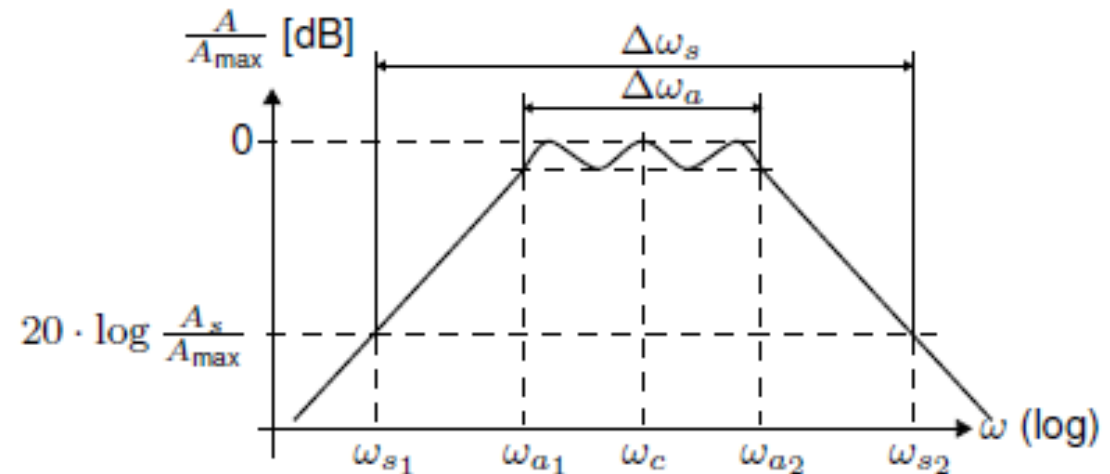
The transfer function of the denormalized **high-pass** filter is (**replace s with s/ω_a in the above**)

$$H(s) = \frac{s^3/\omega_a^3}{(1 + 0,49417/\omega_a s + 0,99421/\omega_a^2 s^2)(1 + 0,49417/\omega_a s)}$$

Low-pass to band-pass transformation

Bandpass filters can be designed from normalized prototype lowpass filters based on the following specification

- The filter function prototype
- The center frequency of the filter ω_c
- The Passband bandwidth $\Delta\omega_a$
- The Stopband bandwidth $\Delta\omega_s$
- The filter's stopband attenuation A_s

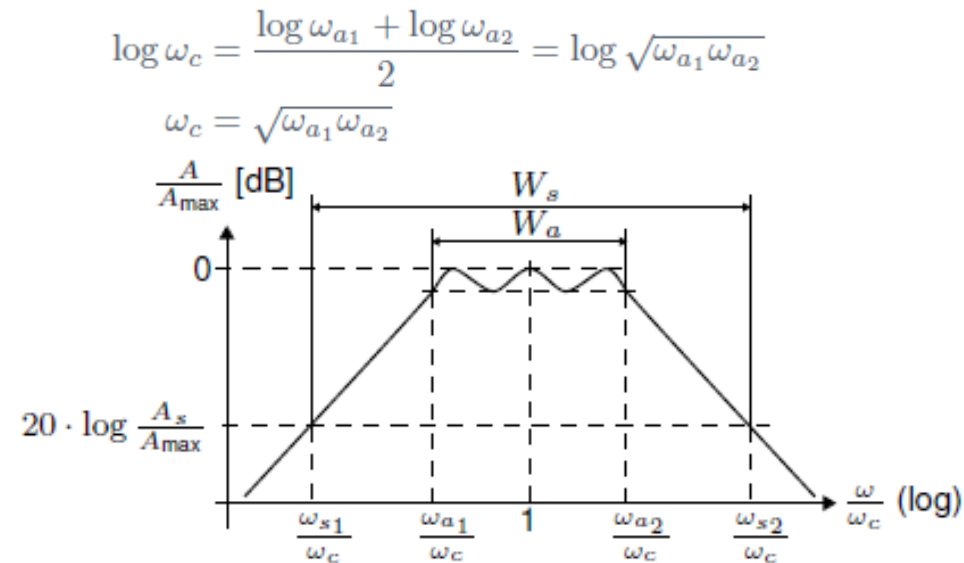


Low-pass to band-pass transformation

The bandpass filter is normalized to the center frequency ω_c , which is midway between ω_{a1} and ω_{a2} on a logarithmic axis, i.e.

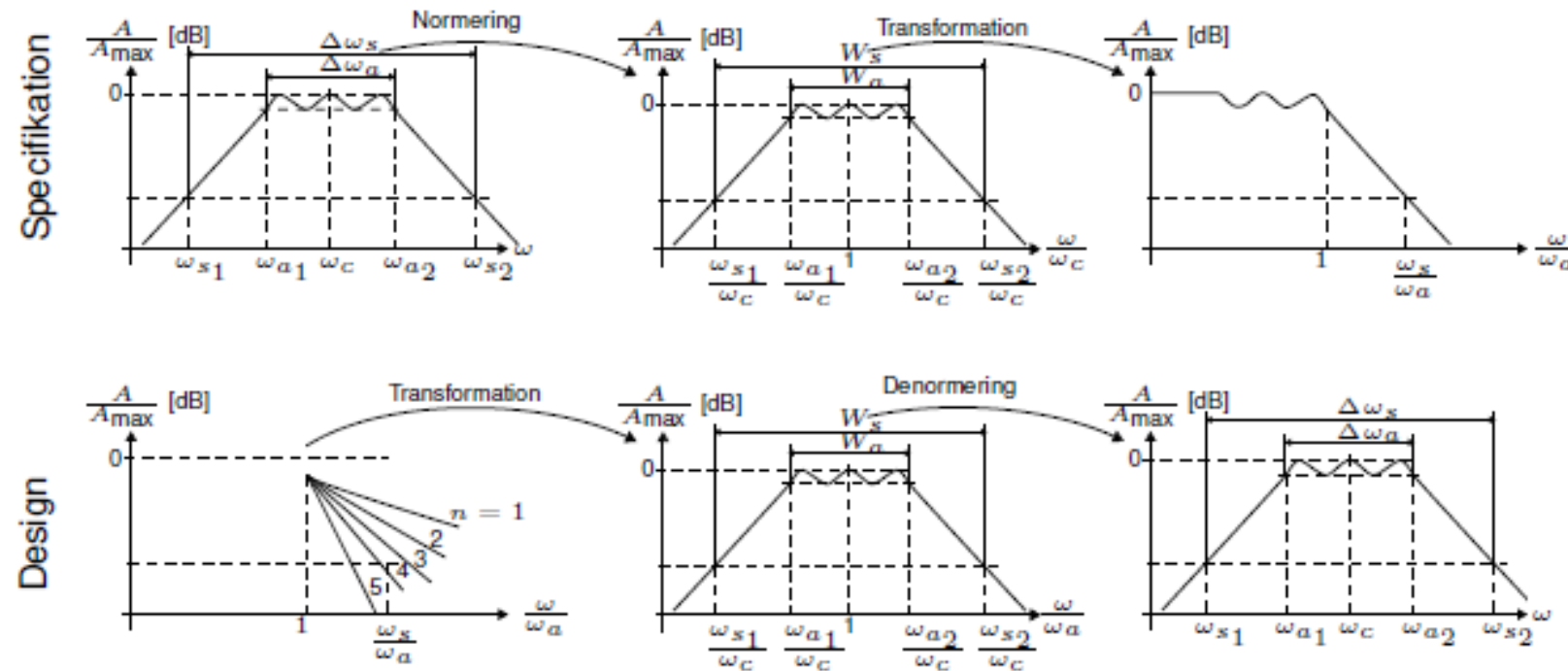
$$W_a = \frac{\Delta\omega_a}{\omega_c}$$

$$W_s = \frac{\Delta\omega_s}{\omega_c}$$



The form factor for a band-pass filter is defined as $F = \frac{W_s}{W_a}$

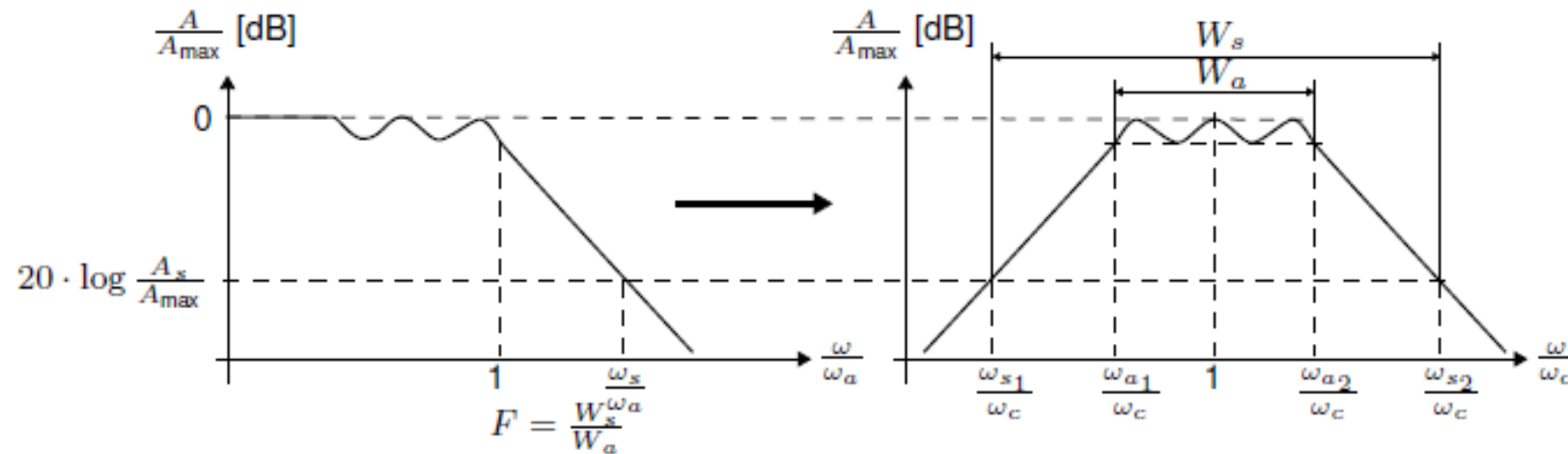
Overview of the design procedure



Filter transformation

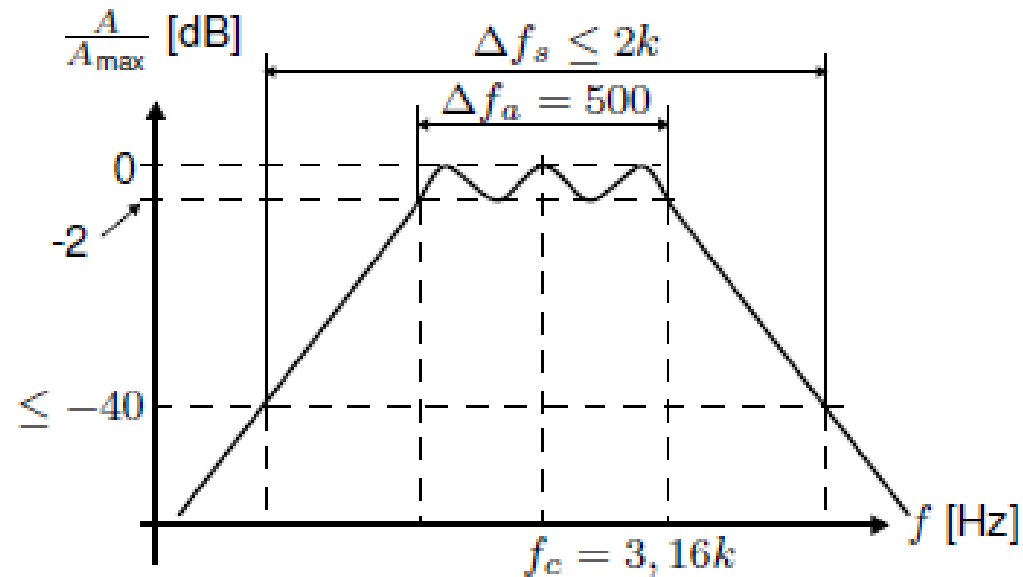
A low-pass filter can be transformed into a band-pass filter by

$$H_{bp}(s) = H_{lp}(\bar{s}) \Big|_{\bar{s} = \frac{1}{W_a} \left(s + \frac{1}{s} \right)}$$



Example: Filter specification

Design a 2 dB Chebyshev bandpass filter that satisfies the following amplitude characteristic.



Example: Low-pass to band-pass transformation

Step 1: Filter specification

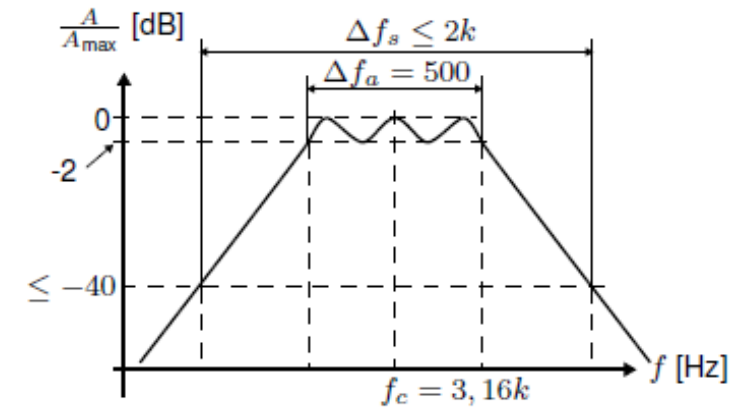
To design the filter, the bandpass filter is normalized, i.e. the standardized stopband width and the standardized passband width are calculated.

$$W_a = \frac{\Delta\omega_a}{\omega_c} = \frac{\Delta f_a}{f_c} = 0.1582$$

$$W_s = \frac{\Delta\omega_s}{\omega_c} = \frac{\Delta f_s}{f_c} = 0.6329$$

Finally, the form factor is calculated

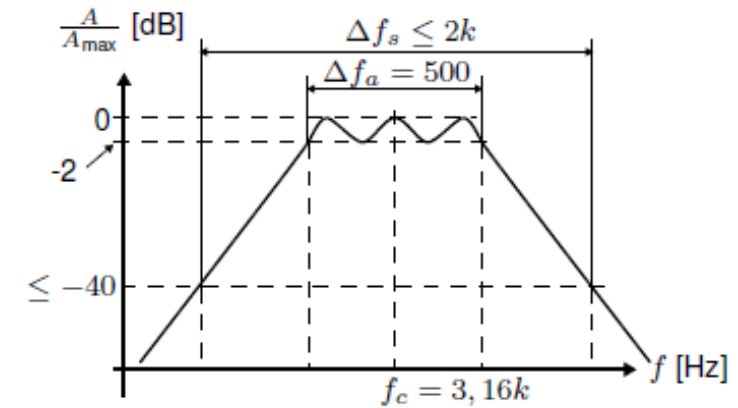
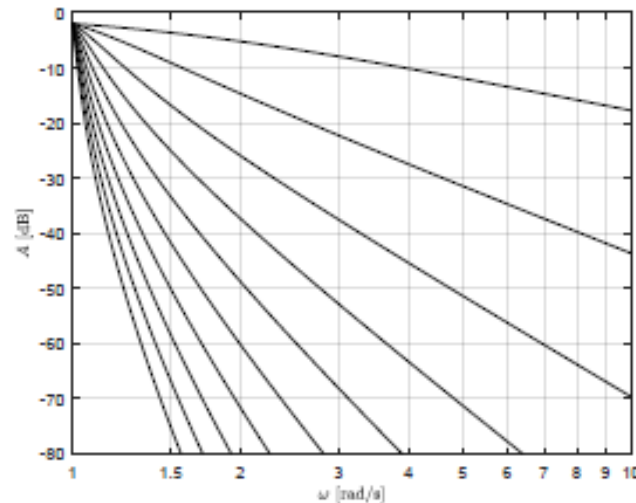
$$F = \frac{W_s}{W_a} = 4$$



Example: Low-pass to band-pass transformation

Step 2: choose filter's order

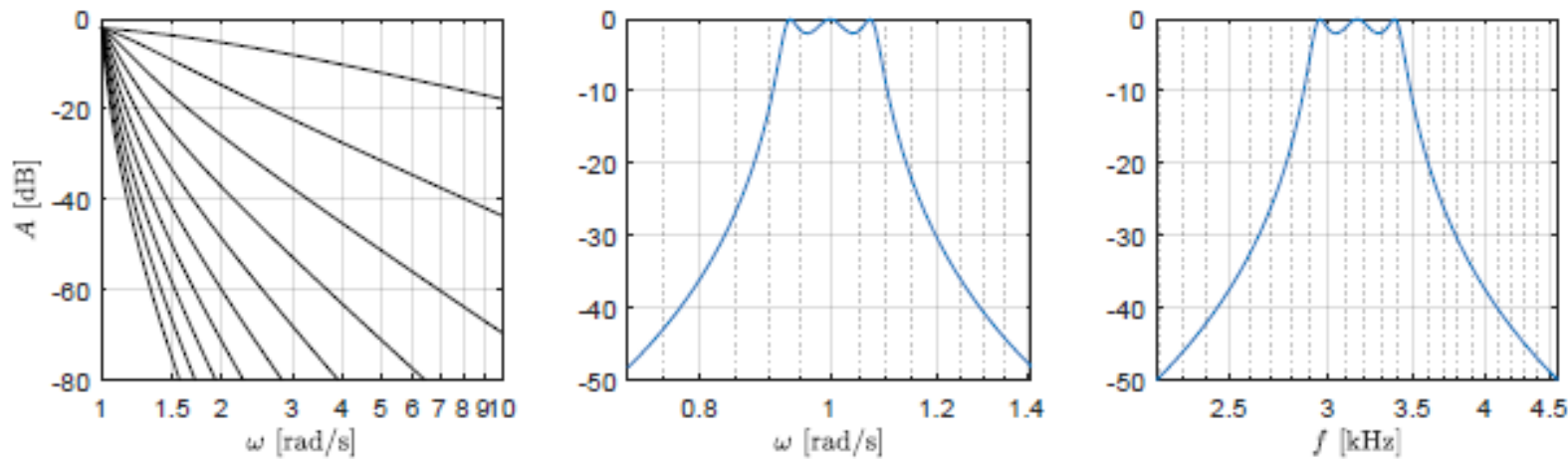
The filter's order is designed based on the amplitude characteristics of 2 dB Chebyshev low-pass filters. It is required that the gain is **below -40 dB** at a frequency of 4, since $F = 4$.



The minimum number for the filter's order that meets the requirement is 3 ($n = 3$).

Example: Low-pass to band-pass transformation

Step 3: Denormalization



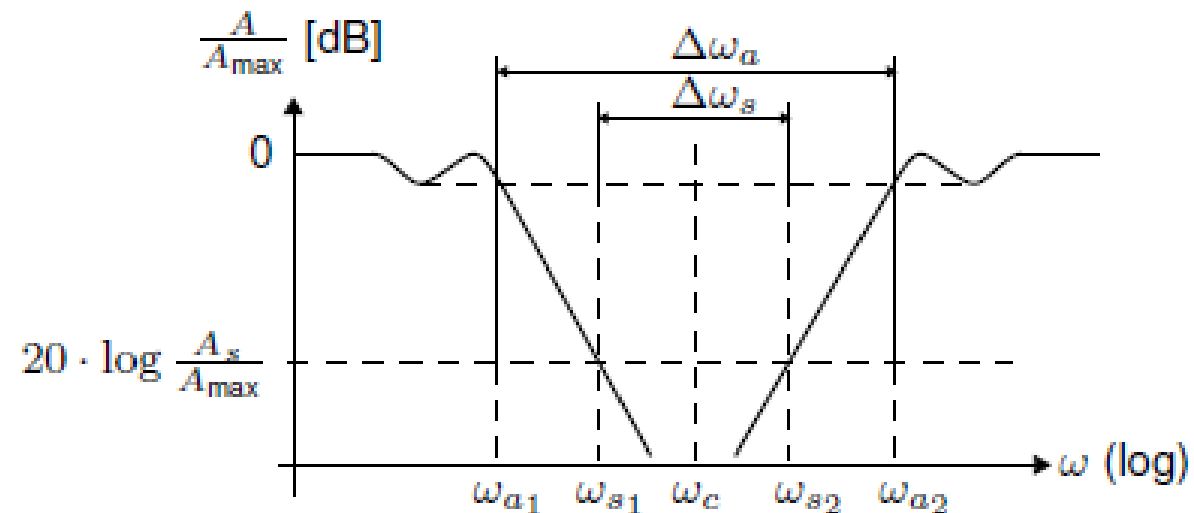
→ The normalized filter is made by **replacing s with $\frac{1}{W_a}(s + \frac{1}{s})$** in the low-pass filter.

→ The denormalized bandpass filter is found by replacing s in the normalized bandpass filter with s/ω_c

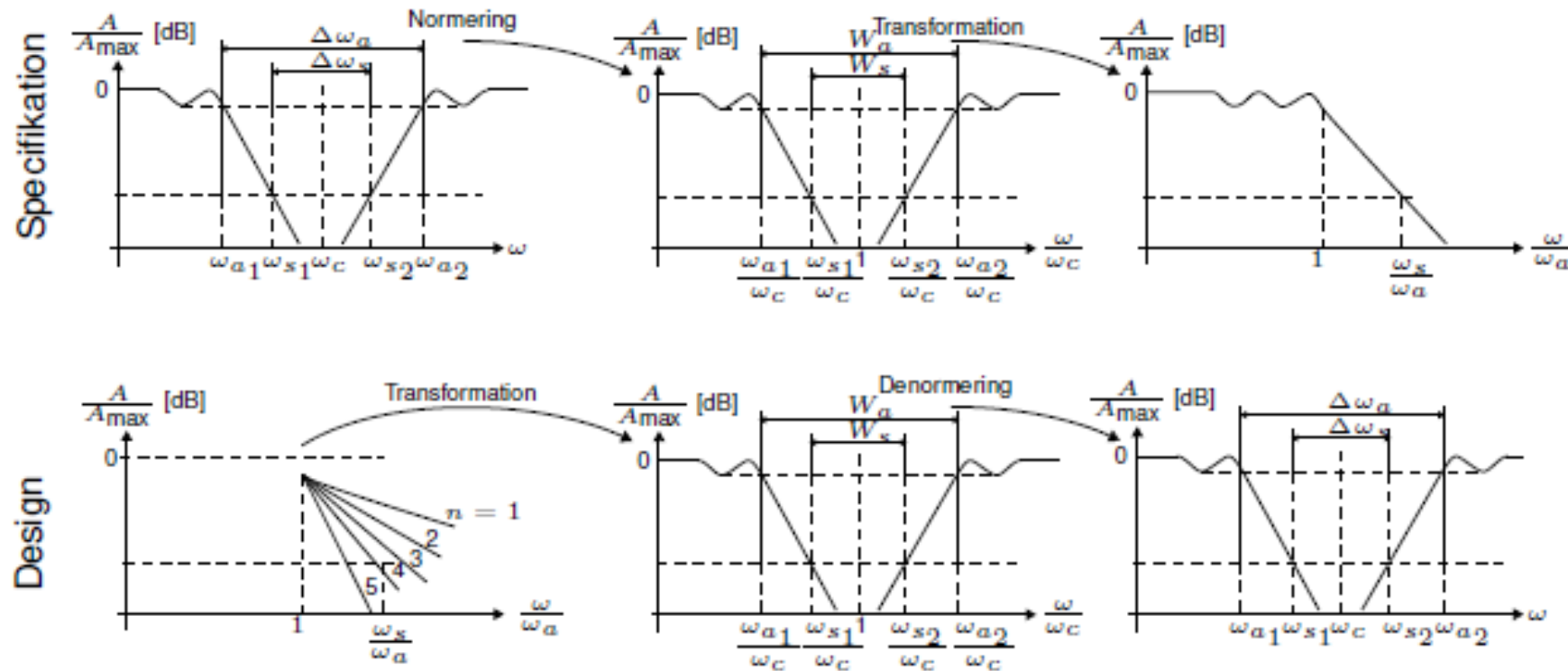
Low-pass to band-stop transformation

Bandstop filters can be designed from normalized prototype low-pass filters based on the following specification

- The Filter function (Bessel, Butterworth, Chebyshev)
- The center frequency of the filter ω_c
- The Passband bandwidth $\Delta\omega_a$
- The Stopband bandwidth $\Delta\omega_s$
- The filter's stopband attenuation A_s



Overview of the design procedure

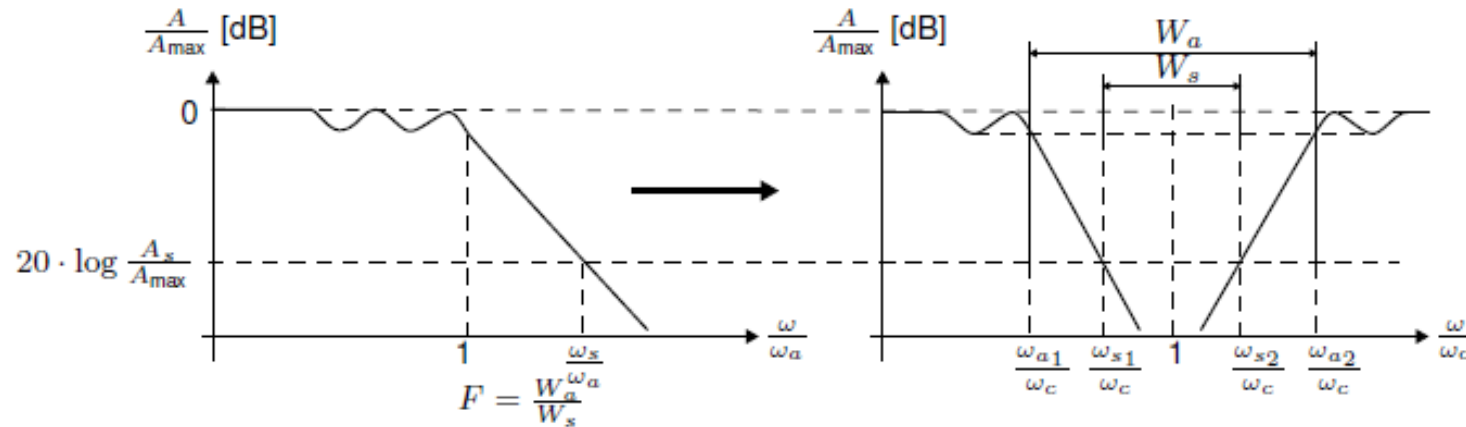


Filter transformation

A low-pass filter can be transformed into a band-stop filter by

$$H_{bs}(s) = H_{lp}(\bar{s}) \bigg|_{\bar{s} = \frac{W_a}{s + \frac{1}{s}}}$$

The transformation is illustrated in the following figure.



The **form factor** of a bandstop filter is defined as $F = \frac{W_a}{W_s}$

Matlab functions

Please check Matlab inherent functions for filter conversion:

lp2lp() %Transform lowpass analog filters to lowpass
lp2hp() %Transform lowpass analog filters to highpass
lp2bp() %Transform lowpass analog filters to bandpass
lp2bs() %Transform lowpass analog filters to bandstop

Method 2

Matlab can directly design the filter function for you!

Tell Matlab the design requirements:

- Filter function (Bessel, Butterworth, Chebyshev)
- Filter order N
- Cutoff frequency of the filter ω_a
- Filter type “s” for analog filter $H(s)$
- Filter type: “low”, “high”, “bandpass”, “stop”
- Other filter parameter such as ripple size R_p

Matlab functions:

- [`\[b,a\] = butter\(n,Wn,ftype\)`](#)
- [`\[b,a\] = cheby1\(n,Rp,Wp,fType\)`](#)
- [`\[b,a\] = cheby2\(n,Rs,Ws,fType\)`](#)
- [`\[b,a\] = ellip\(n,Rp,Rs,Wp,fType\)`](#)
- [`\[b,a\] = besself\(n,Wo\)`](#)

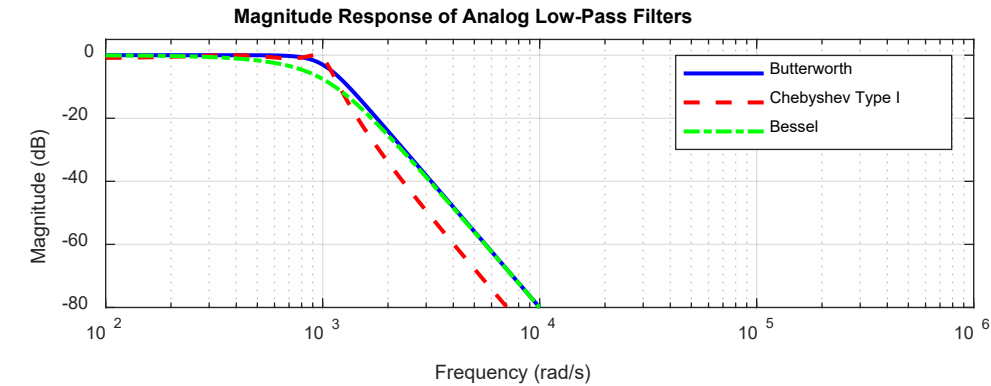
Matlab example

```
% Define filter specifications
N = 4;           % Filter order
wa = 1000;       % Cutoff frequency in rad/s

% Butterworth Filter
[num_butter, den_butter] = butter(N, wa, 's'); % Butterworth filter

% Chebyshev Type I Filter
Rp = 1; % Ripple in the passband (dB)
[num_cheby, den_cheby] = cheby1(N, Rp, wa, 's'); % Chebyshev Type I prototype filter

% Bessel Filter
[num_bessel, den_bessel] = besself(N, wa); % Bessel filter
```



Exercise

6.13. Convert each of the following transfer functions into pole-zero form:

a. $H(z) = \frac{z^2 + 2z + 1}{z^2 + 5z + 6}$

b $H(z) = \frac{1 - 0.16z^{-2}}{1 + 0.7z^{-1} + 0.1z^{-2}}$

c $H(z) = \frac{z^2 + 4z + 5}{z^3 + 2z^2 + 6z}$

6.19. For the digital system

$$y(n) = x(n) - 0.5y(n - 2)$$

with a sampling rate of 8,000 Hz,

- a.** determine the frequency response;
- b.** calculate and plot the magnitude and phase frequency responses;
- c.** determine the filter type based the magnitude frequency response.

Exercise

6.29. Given a filter

$$H(z) = \frac{1 - z^{-1} + z^{-2}}{1 - 0.9z^{-1} + 0.81z^{-2}}$$

- a. plot the magnitude frequency response and phase response using MATLAB;
- b. specify the type of filtering;
- c. find the difference equation;